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# Exploring the Boundaries of Multi-Agent Ensembles: An Evaluation of Aggregation Operators

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## Abstract

An ensemble of language models can be no better than the share of problems that at least one of its members solves. Every way of combining members, which we call an aggregation operator, collects some part of the distance between the best single member and that ceiling. This paper measures that share, on the same data and with the same yardstick, for plurality voting, repeated sampling of one model, multi-round debate, synthesis, selection and certification by judges, routing by an orchestrator, and decomposition into subtasks. The data are a text corpus of 733 problems answered by seven models from five vendors, with every jointly failed problem re-audited and every operator run on it; the same corpus with the frontier models removed; 9,600 public debate traces; the SWE-bench Verified leaderboard [1, 1] matrix of 175 agent submissions on 500 bugs, at the top and in mid-ranked accuracy bands; two public leaderboard matrices; released τ-bench and OS-Harm logs [2]; and a cost-matched study of one model's draws. Most comparisons were pre-registered.

The results divide the operators in two. Operators that only rearrange answers already produced collect nothing: plurality voting is level with or below a best member chosen on held-out problems on every data set, whether the room above the best member is 2 points or 32, because the members fail on the same hard problems; a one-factor model of shared item difficulty reproduces their joint-failure rates and their votes within a few points, the independent Condorcet model over-predicts the vote by up to 27 points where it promises most, and after one difficulty axis is removed the residual correlation among members is zero. Operators that add computation collect a few points, under conditions we can name: repeated sampling pays only where a single draw is below 0.9 and the total budget is about ten times one answer's length; debate collects nothing on a saturated panel, where its apparent gain is the recovery of answers cut off by a token limit, and one to two of four points on a pre-registered pool whose members' errors are unshared, while turning eleven of 146 contested problems all-wrong; a judge panel collects a few points whose sign against the best member depends on the tie rule, and is never clearly above its best judge; decomposition adds nothing on problems a model cannot solve whole. A cheap model's ensemble replaces a stronger model only when the accuracy gap is under about three points and the price ratio is at least the number of copies.

Each result is stated with what it does not show: the text corpus is saturated, four of its models ran under an 8,192-token cap, the label audit was one-sided, several tie rules were chosen in sample, and the agent matrices are leaderboard submissions. The one large prize is the oracle on agent tasks, 10 to 32 points above the best submission, reachable only by a selector with an external verifier. A verifier's two error rates, false accept and false reject, measured per task, predict what a selector collects (v9, v10). On agent tasks the prize goes to a verifier with a small false-accept rate, and neither model judges nor model-written tests, written blind or after reading the code, supply one: a trained critic takes half of the prize at two candidates and a third at sixteen, a judge a fifth, and model-written tests accept 39 to 69% of wrong patches; the best deployable rule found, tests as a gate before a ranking judge, collects 0.28 of the prize at eight candidates.

## 1. Introduction: what this paper does

### 1.1 The question

Several recent systems present a group of language models as something more capable than any one of them: models that vote, models that argue with each other, models that check each other's answers, orchestrators that decide which model should answer, and teams of agents that divide a task among themselves. Their reports typically show the group scoring above the best single model in the group. The question this paper asks is plain: *by how much, on which problems, through which operator, and what sets the limit?*

We use the word **ensemble** for any set of $n$ members whose answers are combined, and **aggregation operator** (or just operator) for the rule that combines them. The operators studied here are listed in Section 2. They differ in when they act and what they see: voting and judging act after all answers exist; routing acts before any answer exists; debate lets members see each other's work and revise; decomposition changes the problems the members are asked to solve. Despite these differences they share one bound, which gives the paper its structure.

### 1.2 The bound that organises the paper

Call the **oracle** the share of problems that at least one member answers correctly. A selector that could always pick a correct answer when one exists would reach exactly the oracle; no selector, judge or router that returns one of the members' answers can do better. The **best member** is the member with the highest accuracy. The distance from the best member to the oracle is therefore the whole prize that any selection-type operator can win. Voting is bounded differently, by the number of effectively independent votes the members carry, but on every data set here that number turns out to be small, and in practice the vote never reaches the oracle either.

So instead of asking "does the ensemble beat the best model" we ask two questions for every operator: how large is the distance from the best member to the oracle on this data, and what fraction of that distance does the operator collect? The first question depends only on the data; the second is the operator's merit. Keeping them apart avoids a confusion that runs through much of the literature, where a small gain is read as a weak operator when it is in fact a saturated benchmark, or a large gain is read as a strong operator when it is in fact a weak baseline. It also tells us where to look: a result on a saturated pool says little, so for every operator we also measured a pool with room above its best member.

### 1.3 What we measured

The measurements come from seven data sets, chosen so that each operator can be seen on problems with and without room above the best member:

- A **text corpus** of 733 problems from four sets (MMLU-Pro, GPQA Diamond, GSM-Hard, olympiad mathematics [3, 4, 5, 6, 7]) answered by seven models from five vendors, with repeated samples, debates, judge panels, personas and decomposition run on the same problems, and every problem on which all seven models were scored wrong re-examined by hand. This is our own data.
- The **same corpus with the frontier models removed**: voting re-counted for the four non-frontier members, and a pre-registered debate, synthesis and judge run among them on the problems where they disagree.
- **Public debate traces**: 9,600 three-agent debate transcripts released by other authors, 16 configurations.
- The **SWE-bench Verified leaderboard [1, 1]**: per-bug outcomes of 175 agent submissions on the same 500 bugs, read at the top and in mid-ranked accuracy bands. These are leaderboard entries, not runs we made.
- Four **public correctness matrices** of 28 to 67 language models (three released by Chen et al., one from the Open LLM Leaderboard [8]); two of them, 52 models on MMLU-Pro and 28 on MMLU, hold

enough members of equal, mid-ranked accuracy to isolate a band.
- **τ-bench trial logs [9, 10]** and **OS-Harm logs [2]**, released with the benchmarks: the same agent run several times, and five computer-use agents on 150 safety tasks.
- A **cost-matched study**: sixteen draws of one model on about sixty problems per set, scored at every token budget, with nominal prices attached.

Only the first, second and last involved model calls of our own. Everything on the agent benchmarks and the public matrices is computed from released outcome matrices, with no trajectories read. Where we could, we wrote the hypothesis and the decision rule before looking at the data; Section 4 says which results were pre-registered and which were decided afterwards, and which pre-registered predictions failed.

### 1.4 What we found, in one paragraph

Operators that only rearrange answers already produced collect nothing. Plurality voting is level with or below a best member chosen on held-out problems on every data set: on the saturated text corpus, where the room above the best member is at most 2.3 points; on the same corpus without its frontier models, where the room is 1 to 9 points; among mid-ranked SWE-bench submissions and in the matched-accuracy bands of two leaderboard matrices, where the room is 14 to 32 points. The reason is the same everywhere: the members' errors share one latent axis, the difficulty of the item, and almost nothing else, so three to seven members vote like one or two; a one-factor model reproduces their joint failures and their votes, and the independent Condorcet model over-predicts the vote by up to 27 points. Operators that add computation collect a few points, under conditions we can name. Repeated sampling pays on hard problems at about ten times one answer's tokens. Debate collects nothing on a saturated panel, where its apparent gain is the recovery of answers cut off by a token limit, and one to two of four points on a pool whose members' errors are unshared, at the price of turning some problems all-wrong. A judge panel collects a few points whose sign against the best member is set by the tie rule. Decomposition adds nothing on problems a model cannot solve whole. The prize nobody collects is the oracle on agent tasks, 10 to 32 points above the best submission, which only a selector with an external verifier could reach.

### 1.5 Contributions

1. One yardstick, the share of the best-to-oracle distance collected, applied to seven operators on one audited corpus and on four public data sets, with the best member cross-validated.
2. The finding that the null for voting is not a ceiling effect: it holds with 1 to 9 points of room in non-frontier pools and 14 to 32 points in mid-ranked bands, exactly where the independent Condorcet model promises the most and fails by up to 27 points.
3. Evidence from three independent methods that one difficulty factor accounts for almost all shared error among models and among agent submissions, which bounds every rearranging operator at once.
4. A label audit showing that 45 of the 48 text problems on which all seven models were scored wrong were defects of the benchmark; after correction the seven fail together on 1 problem in 733.
5. Debate taken apart: a self-only control, truncation accounting, and a pre-registered run on a pool with unshared errors whose four predictions all missed toward a larger effect.
6. A break-even rule for replacing an expensive model by an ensemble of a cheaper one at equal spend.
7. Pre-registered co-failure measurements on agents with tools and on safety breaches, from released data at zero cost.

Section 2 introduces the terms in words and then as formulas. Section 3 is the results: a ledger of what each operator collects, operator by operator, then the explanation and what is new. Section 4 describes the data and the protocol. Sections 5 to 9 treat the operators (voting and self-consistency, debate, judges and certification, routing, decomposition); Section 10 prices them at equal budget; Section 11 reads them on agents with tools

and on safety. Section 12 places the results in the literature, Section 13 lists what the data do not show, and Section 14 concludes.

## 2. Terms, first in words, then as formulas

### 2.1 In words

A **member** is one language model, or one model wrapped in a tool-using scaffold, that returns an answer to a problem. An **ensemble** is a set of members together with an **operator** that turns their answers into one final answer. We study seven operators:

- **Plurality voting.** Each member answers once; the answer given by most members wins. "Majority" is the same thing when members are counted as right or wrong rather than by answer identity.
- **Self-consistency.** One member answers $k$ times at non-zero temperature; the most frequent answer wins. It is voting with one model as every voter.
- **Debate.** Members answer, then see each other's answers and revise, for several rounds; the final round is voted. Unlike voting, each member's later answer depends on the others.
- **Selection by a judge** (best-of-$n$ with a posterior check). After the members answer, a judge, itself a model, picks one of the candidate answers. A **panel** is several judges whose picks are voted.
- **Certification.** A judge says of one candidate whether it is correct. Unlike selection it can reject every candidate.
- **Routing** (best-of-$n$ with a prior check). An orchestrator reads the problem and sends it to one member before any answer exists; a multi-turn orchestrator may change member between turns.
- **Decomposition.** The problem is split into subtasks, each solved separately, by the same member or by different ones, and the parts are assembled.

Two further operators appear in Sections 6 and 7. In **synthesis** one call reads all members' answers and writes a single new answer; unlike selection it may produce an answer no member gave. A **partial verifier** is a check with a known error rate that is not a model's opinion: a test suite, a compiler, program execution. We also use these words throughout: the **prize** is oracle minus best member, in points; a problem is **contested** when at least one member is right and one wrong; an accuracy **band** is the set of members whose accuracy falls in a stated range; a **tie rule** is what the vote does when two answers have equal support (we report three: the first member in a fixed order wins, ties split evenly, the strongest member wins); the **modal-hit rate** is the share of problems on which a model's most frequent answer is correct; an answer's **natural length** is the tokens a model uses when not cut off; the **union gain** of an operator is how many problems it solves that none of the baseline attempts solved.

The **best member** is the member with the highest accuracy on the evaluation problems. When it is identified on the same problems it is then scored on we say it is chosen *in sample*; this flatters it, because the maximum of several noisy accuracies is biased upward. When it is chosen on other problems (a held-out half, or fixed in advance) we say *held-out*. Every "gain over the best member" in this paper states which one is meant. Operators that spend no more than their members (voting, routing) are compared with the held-out best member as a share of the prize; operators that spend more (repeated draws, debate, judges, synthesis) are compared with the single answer or the round-0 vote in points, and charged for their tokens in Section 10.

The **oracle** is the share of problems that at least one member answers correctly; equivalently, one minus the **joint failure rate**, the share of problems every member gets wrong. The oracle is what a perfect selector would score and is therefore the ceiling of selection, certification and single-turn routing. It is also called *coverage* in the agent literature and *pass@n* when the members are draws of one model.

Two members are **correlated** when they tend to be right and wrong on the same problems. We measure this on the 0/1 correctness vectors with the ordinary correlation coefficient, written $\varphi$, and with the **tetrachoric** correlation $\rho$, which asks what correlation between two latent continuous variables would produce the observed 2×2 table of right/wrong pairs. The tetrachoric value is the natural quantity in the one-factor model below; it is larger than $\varphi$ whenever accuracies are high, and the quick approximation we use for it is biased upward in that regime, so we read it as "much larger than $\varphi$" rather than as a precise number.

The **effective number of independent votes**, $n_{\text{eff}}$, is the number of independent voters whose majority would have the same variance as the actual $n$ correlated voters. It is a summary of $\varphi$, not a measure of how similar the models are in any other sense, and it bounds what voting can add: when $n_{\text{eff}}$ is near 1, the vote behaves like one member.

The **token cap** is the maximum length a member may write. An answer cut off by the cap is *truncated*. Whether a truncated answer counts as a wrong vote or as a missing vote changes several results; we say which rule is used. **Equal budget** means that $n$ draws of $B/n$ tokens each are compared with one draw of $B$ tokens.

A result is **pre-registered** when its hypothesis, the quantity to compute and the rule for calling it supported or not were written and committed before the data were looked at.

## 2.2 As formulas

Let there be $n$ members and $N$ problems, and let $x_{ij} = 1$ if member $i$ is correct on problem $j$, 0 otherwise. Member accuracy is $p_i = (1/N)\,\Sigma_j\, x_{ij}$.

$$\text{best} = \max_i p_i; \quad \text{oracle} = (1/N)\,\Sigma_j \max_i x_{ij}; \quad \text{joint failure} = 1 - \text{oracle}.$$

For any operator $A$ that returns one of the members' answers (selection, certification, single-turn routing):

$$\text{acc}(A) \le \text{oracle}, \quad \text{and the } \textbf{collected share} \text{ is } \; s(A) = (\text{acc}(A) - \text{best}) \,/\, (\text{oracle} - \text{best}).$$

The lower end is not guaranteed: a judge can pick worse than the best member, so $s$ can be negative. Selection beats the best member exactly when the judge's rate of correct picks on problems where the best member is wrong and another member is right exceeds its rate of wrong picks on problems where the best member is right.

**Correlation and effective votes.** With $\varphi_{ik}$ the Pearson correlation of the vectors $x_{i\cdot}$ and $x_{k\cdot}$ and $\bar{\varphi}$ their mean over pairs, Kish's design effect [11] gives

$$n_{\text{eff}} = n \,/\, (1 + (n - 1)\,\bar{\varphi}).$$

Seven members with $\bar{\varphi} = 0.40$ carry $n_{\text{eff}} = 2.06$; with $\bar{\varphi} = 0.68$ they carry 1.38.

**Independent Condorcet.** If $n$ voters are independent and each is correct with probability $p$, the majority is correct with probability

$$P_{\text{maj}}(p, n) = \Sigma_{k > n/2}\, \mathrm{C}(n, k)\, p^k\, (1 - p)^{n-k},$$

which exceeds $p$ whenever $p > ½$ and grows with $n$. With unequal $p_i$ the sum runs over subsets; the gain over the best member is then a non-linear function of the whole accuracy profile, which is why linear "diversity" scores predict it badly (Section 5). The Kish prediction used in Section 5 evaluates $P_{\text{maj}}$ at $n = n_{\text{eff}}$ (continuous, normal approximation) and at the mean member accuracy.

**Correlated Condorcet with one difficulty factor.** Let each problem have a latent difficulty $D_j \sim N(0, 1)$ and each member a latent score $Z_{ij} = (\sqrt{\rho}) \cdot D_j + (\sqrt{(1-\rho)}) \cdot E_{ij}$, with $E$ independent $N(0, 1)$; member $i$ is correct when $Z_{ij}$ exceeds a threshold $t_i$ chosen so that $P(\text{correct}) = p_i$. Here $\rho$ is the tetrachoric correlation between any two members. This Gaussian copula predicts the joint failure rate and the majority rate from ($p_1$, …, $p_n$, $\rho$) alone. When its predictions match the observed rates, one shared difficulty axis is enough to explain the members' agreement.

**Eigenvalue count.** If the $n \times n$ correlation matrix of the members' correctness were pure noise, its eigenvalues would lie below an edge given by the Marchenko–Pastur law [12] for a matrix of that shape. Each eigenvalue above the edge is one shared axis of variation. One eigenvalue above the edge means one common factor, which in this setting is item difficulty; more would mean additional structure, such as families or scaffolds, that voting or selection could exploit.

**Item-response residuals.** The two-parameter logistic model fits $P(x_{ij} = 1) = \sigma(a_j(\theta_i - b_j))$ with member ability $\theta$, item difficulty $b$ and item discrimination $a$. The **residual correlation** between members is the correlation of $x_{ij} - \hat{P}_{ij}$; it is what remains of their agreement after one difficulty axis. The **family excess** is the mean residual correlation within a vendor's models minus that between vendors.

**Equal-budget break-even.** Let $p(B)$ be a model's accuracy when allowed $B$ tokens. Voting with $n$ draws at equal budget pays iff

$$P_{\text{maj}}(p(B/n), n_{\text{eff}}) > p(B),$$

and selection with a verifier pays iff acc(selector) > $p(B)$ and the candidates' cost plus the verifier's cost is at most the single model's, where acc(selector) is given by the next paragraph.

**Selection with an imperfect verifier.** A verifier accepts a wrong candidate with probability $\alpha$ (false accept) and rejects a correct one with probability $\beta$ (false reject). Under the threshold rule the selector picks uniformly among accepted candidates, and uniformly among all if none is accepted. On a problem with $k$ correct candidates out of $n$, let $A_r$ be the number of correct candidates accepted and $A_w$ the number of wrong ones accepted:

$$A_r \sim \text{Binomial}(k, 1-\beta),\ A_w \sim \text{Binomial}(n-k, \alpha),$$

$$\text{acc}(k, n, \alpha, \beta) = E[\ A_r / (A_r + A_w)\ ]\ (k/n \text{ when nothing is accepted}),$$

(the chance that the candidate drawn from the accepted pool is correct), and the selector's accuracy on a benchmark is the mean of acc($k_i$, $n$, $\alpha$, $\beta$) over problems with their observed $k_i$, which is where the shared failures enter. As $n$ grows with the base rate $p = k/n$ fixed, the per-problem accuracy tends to the verifier's precision, $[p(1-\beta)] / [p(1-\beta) + (1-p)\alpha]$, not to 1 [13]: with $\alpha > 0$ the selector never reaches the oracle, and the loss is largest on hard problems, where the prize sits. A false reject loses a problem only if all $k$ correct candidates are rejected, so more candidates heal it; a false accept adds wrong candidates in proportion to $n - k$, so more candidates hurt. In data the rates vary by problem, ($\alpha_i$, $\beta_i$), and the formula is used with per-problem rates (Section 7.7). The single-rate bound of earlier drafts, $\text{acc(selection)} \le (1 - P_{\text{all wrong}}) \cdot (1 - \varepsilon)$, merged $\alpha$ with $\beta$ and gave every problem the same rate; it is superseded. In dollars, with price ratio $r$ between the expensive and the cheap model, the cheap ensemble of $n$ draws costs $n/r$ of one expensive answer; it is worth considering only if $r \ge n$ and the accuracy gap is smaller than what $n$ correlated draws can add (Section 10).

## 3. Results: what each operator collects, and when

## 3.1 The frame

Every operator in this paper returns, in the end, an answer that some member produced, or an answer built from members' answers. Its ceiling is therefore the **oracle**, the share of problems at least one member solves, and the whole prize it can win is the distance from the **best member** to that oracle (Section 2). For each operator and each data set we report two numbers: the prize, oracle minus best member, which depends only on the data; and the share of the prize the operator collects, which is the operator's merit. The best member is chosen on held-out problems wherever the data allow, because a best member chosen on the scored problems is flattered by selection.

One distinction organises what follows. Operators that only *rearrange* answers that already exist (plurality voting; a router that picks one member in advance) collect nothing, or at most a small minority of the prize, on any of our data sets: voting nothing over a held-out best member, routers a tenth to a quarter of their pool's gap by their own tables and by our arithmetic. The reason is that the members fail on the same hard problems: their correctness shares one latent axis, item difficulty, and almost nothing else, so three to seven members vote like one or two. Operators that *add computation* (repeated draws of one model, debate, a judge call, a synthesis call) collect a few points, and only where the single answer is weak and the members' errors are not already shared; they pay several times the tokens, and debate loses part of the coverage it was meant to collect. The two kinds are compared against different baselines, on purpose: a rearranging operator spends no more than its members, so its merit is its share of the prize over a held-out best member; an operator that adds computation spends more, so we report its gain in points over the single answer or the round-0 vote and then charge it for the extra tokens in Section 10. The one large prize is the oracle on agent tasks, 10 to 32 points above the best submission; Section 7.7 measures on released and on our own candidates what an imperfect verifier collects of it: a model verifier half at two candidates and a third at sixteen, model-written tests at most a quarter, tests as a gate before a judge 0.28.

## 3.2 The ledger

**Table 3.1. What each operator collects. Prize = oracle − best member on that data set, in points. "Collects" is the operator's gain over the best member (held-out where marked) as a share of the prize, or the gain in points where no oracle applies. Evidence: PR = pre-registered; IS = a choice made in sample (tie rule, best member, pool) that the number depends on; CI = an interval is reported.**

| Operator | Data set | Prize | Collects | Pays when | Cost | Evidence | Section |
|---|---|---|---|---|---|---|---|
| Plurality voting | Text corpus, 7 models, 733 problems | 0.0–2.3 | ≈ 0: level with a held-out best (split-half +1.8 [0.0, +3.3] for the best on MMLU-Pro, ≤ 0 elsewhere; Opus fixed 714 vs 708, $p = 0.26$) | never over a held-out best; insurance against a fixed mid-ranked member | 7 answers | CI, IS (pool) | 5.3 |
| | Non-frontier pools (4 / 3 models), same problems | 1.1–8.8 | ≤ 0; where one member dominates (GPQA, GLM 0.989 vs 0.76–0.87) the vote loses 8.9 [4.3, 15.2] | as above; beats DeepSeek fixed in advance (10/33, $p = 0.001$) | 3–4 answers | CI, IS (pool chosen after Table 5.1) | 5.4 |

| Operator | Data set | Prize | Collects | Pays when | Cost | Evidence | Section |
|---|---|---|---|---|---|---|---|
| | SWE-bench bands, acc 0.4–0.7 (28–31 submissions) | 19–32 | < 0: majority 1.6–4.4 below best; random panels of 7 beat their best in 1–17 % | never | band | CI, IS (band) | 5.5, 11.2 |
| | SWE-bench top 7 | 9.8 | 0.02 (0.794 vs 0.792; 12 won / 11 lost) | never | 7 runs | PR, CI | 11.2 |
| | Chen MMLU-Pro and Leaderboard MMLU, acc 0.5–0.7 bands | 14–25 | ≈ 0 (held-out majority within 0.3 of the best, beats it in 45–49 % of subsets) / < 0 (4–6 below, < 1 %) | never | 3–5 answers | CI, held-out best | 5.5 |
| Self-consistency (k draws of one model) | Text corpus, k 8–16, ~60 problems per set | — | +4.3 / +6.7 on GPQA; +14.5 on olympiad (49 of 60 problems; +6.1 vs the first draw); −0.6… +1.7 on MMLU-Pro, GSM-Hard | single draw < 0.9, hard problems | 8–16× | no CI | 5.7 |
| | Equal tokens | — | loses 0.3–31 at every budget one draw can use; +7…+12 on olympiad at 32,768 tokens; +4.3 on GSM-Hard at 8,192 (n = 3) | budget ≈ 10× an answer's natural length and single < 0.9 | equal | PR, CI | 10.3 |
| | Dollars (Haiku vs Opus) | — | reaches Opus only on MMLU-Pro at k = 15 (10× price) and GSM-Hard at k = 5 (3.4×); ≤ 2.7 points from nine copies at the measured draw correlation | accuracy gap < ~3 points and price ratio ≥ n ≥ 3 | 3–15× | PR, model-based bound | 10.5–10.6 |
| Debate (3 rounds) | Text corpus, 6 models, 225 problems | — | +4.9 or +2.2 by tie rule; 10 of 11 fixes on cap-truncated answers; 1 fix on the 162 complete problems; six copies +3.5, half | only where an agent had not finished, or as "think again" | 3× rounds, 6 agents | IS (tie, stop rule), no CI, $p = 0.06$ | 6.3 |

| Operator | Data set | Prize | Collects | Pays when | Cost | Evidence | Section |
|---|---|---|---|---|---|---|---|
| | | | of it with no peers | | | | |
| | Non-frontier pool, 146 contested + 40 control + 10 all-wrong | 3.6 | ≈ 0.36 (0.947 → 0.963 vs best 0.950, $p$ = 0.052); contested 0.801 → 0.870 (14/4, $p$ = 0.031); coverage 1.000 → 0.925 | members' errors unshared ($\bar{\phi}$ −0.10 at round 0) | 12 agent-turns per problem | PR (all four predictions missed, toward more effect); discordant counts, no interval | 6.4 |
| | Public traces, 16 files | — | +6…+20 over the round-0 modal-hit rate on weak models; "31–45 % of all-wrong corrected" on 2 of 16 files | weak models | 3 agents × 3 rounds | PR (ICC), CI | 6.5 |
| Synthesis (one call reads n answers) | Text corpus / non-frontier pool | — | 0 of 16 all-wrong recovered (saturated pool); 0.849 on the 146 contested (debate 0.870), 1 of 10 all-wrong with a novel answer | as a cheap selector | 1 call | PR (missed) | 6.3, 6.4 |
| Selection by a judge panel | Text corpus, 5 judges, 74–75 contested | — | +8.1 / +4.0 / −1.4 over the solvers' vote by tie rule; whole set 0.940–0.953 vs best solver 0.945 | judge is a better solver reading the candidates | 5 calls | IS (tie, best), $p$ = 0.21 | 7.4 |
| | Non-frontier pool, 3 judges, 146 contested | 3.6 | +3.4 / +6.8 / +5.5 over the vote; 733: 0.939–0.958 vs best 0.950; not above its best judge | as above | 3 calls | PR (±3 missed), $p$ = 0.17 | 7.5 |
| Certification | 54 wrong candidates | — | only unanimity of five keeps false accepts under 5 % (1 of 54, [0.3 %, 9.8 %]); without Sonnet and Qwen (unanimity of the other three) 8 of 54 | one strict judge | 5 calls | CI | 7.6 |

| Operator | Data set | Prize | Collects | Pays when | Cost | Evidence | Section |
|---|---|---|---|---|---|---|---|
| Routing (prior choice) | Text corpus; published systems | ≤ 2.4 (ours) | per-set route 0.7 points (≈ 0.27, in sample); Trinity ≈ 0.15 of its gap; Chen ≈ 0.09; Conductor, Fugu: no oracle reported | pool selected for complementarity | 1 answer + router | not measured by us | 8 |
| Decomposition (equal budget) | 97 problems, two models | — | 0.00 [−0.29, +0.29] on 14 hard numeric (0 of 7 never-solved); +0.34 [+0.18, +0.50] where one pass is cut off; −0.41 on easy problems; team ≤ split | a model that runs out of tokens | 3.5–9 draws (DeepSeek) | PR, CI; mechanism not isolated | 9 |
| Co-failure of agents (not an operator) | τ-bench reruns; OS-Harm safety breaches | — | one agent rerun 4× = 1.4–1.9 independent runs; across-model ϕ 0.29–0.47; misuse breaches shared (ρ 0.67, $n_{eff}$ 1.4), injection weakly shared (ρ 0.28, permutation $p$ = 0.0005); breach vs task-error difference not significant | — | — | PR; 5 agents on OS-Harm | 11.3–11.4 |
| Oracle / partial verifier | SWE-bench, τ-bench; R2E-Gym and SWE-rebench candidates | 10–32; +14…+25 (pass@4); 32.5 (R2E), 12.6 (SWE-rebench) | model verifier (AUROC 0.805): 0.51 [0.46, 0.56] of the prize at n=2, 0.36 [0.29, 0.42] at n=16; agent's own tests (α 0.69): 0.02–0.04; our judge 0.15–0.19, our reading tests 0.23, tests then judge 0.28 at n=8 on 200 contested issues; 82 selector-flag submissions indistinguishable from peers | external verifier with small α | n runs + verifier | measured on released candidates, no calls | 7.7 |

### 3.3 Operator by operator

**Plurality voting collects nothing, on any data set, over a best member chosen out of sample.** On the text corpus the best member scores 0.975 to 1.000 and the oracle 0.998 to 1.000, so there is almost nothing to collect; the in-sample best leads the vote by 0 to 1.9 points, the split-half best by +1.8 [0.0, +3.3] on MMLU-Pro and by nothing elsewhere, and Opus fixed in advance scores 714 of 733 against the vote's 708 ($p$ = 0.26) . Removing the frontier models opens 1 to 9 points of room and the vote collects none; where one member dominates its pool the vote loses to it by 8.9 points with an interval clear of zero. Among mid-ranked SWE-bench submissions at accuracy 0.4 to 0.7, with 19 to 32 points of room and 58 to 93 bugs nobody in the band resolves, the band majority is 1.6 to 4.4 points below the best member and random panels of seven beat their own best in 1 to 17 per cent of draws; in the matched-accuracy bands of two public leaderboard matrices the held-out majority ties the held-out best or trails it by 4 to 6 points. What the vote does give is insurance: the plurality of three to five mid-ranked members scores about where the best of them scores, without knowing in advance which that is, and beats a member fixed in advance (DeepSeek wins 10 problems against the vote and loses 33, $p$ = 0.001).

**Self-consistency pays on hard problems and nowhere else, and at equal tokens it loses.** Eight to sixteen draws of one model gain +4.3 to +6.7 points on GPQA Diamond and up to +14.5 on the olympiad set (on the 49 problems where every draw finished; +6.1 over the first draw), and −0.6 to +1.7 on the two easier sets, with no intervals. Given the same total tokens as one answer, splitting them among voters loses by 0.3 to 31 points at every budget one draw can actually use, and pays only when the budget is about ten times an answer's natural length and the single draw is below 0.9 (olympiad +7 to +12 at 32,768 tokens). In dollars a reasoning Haiku answer costs 0.7 to 6.6 Opus answers, and its ensemble reaches Opus only on MMLU-Pro at fifteen draws (ten times the price) and on GSM-Hard at five; at the measured draw correlation nine copies add at most 2.7 points, so a cheap ensemble can stand in for a stronger model only when the gap is under about three points and the price ratio is at least the number of copies.

**Debate collects one to two points where the members have something to exchange, zero where they do not, and always at a loss of coverage.** On the saturated six-model panel the measured gain, +4.9 or +2.2 points by tie rule, sits on answers an agent had not finished within its token cap (10 of 11 fixes) and on "thinking again" (the self-only control fixes 8 and breaks 4 with no peers); on the 162 problems every agent answered it fixes one. On a pool of four non-frontier models whose errors on the contested problems are uncorrelated ($\bar{\varphi}$ −0.10), a pre-registered three-round debate moves those problems from 0.801 to 0.870 (14 fixed / 4 broken, $p$ = 0.031) and the corpus from 0.947 to 0.963 against the best member's 0.950 ($p$ = 0.052): about 1.3 of the 3.6 points to the oracle. It also turns eleven contested problems all-wrong, nine of them unanimously, so the oracle over the four falls from 1.000 to 0.925, and a correct agent facing three disagreeing peers abandons its answer 46 per cent of the time. All four pre-registered bounds for that run were missed, each toward a larger effect than the saturated pool had suggested. In 9,600 public traces debate raises within-problem agreement to 0.62–0.99 and leaves about one effective voter by round 3, while on weak models it exceeds the round-0 modal-hit rate by 6 to 20 points.

**A judge panel collects a few points over the solvers' vote; the sign against the best member is set by the tie rule, and the panel is never clearly above its best judge.** Five judges lead the solvers' plurality by +8.1, +4.0 or −1.4 points depending on how ties are broken, and over the whole set land within a point of the best solver; three non-frontier judges over the non-frontier candidates lead the vote by +3.4 to +6.8 and land between 0.939 and 0.958 against the best member's 0.950; on the contested problems the panel's 0.856 is within noise of its best judge alone (DeepSeek picks 0.829). Certification is safe only by unanimity of five (1 of 54 wrong accepted, Wilson [0.3 %, 9.8 %]), and that rests on the two strictest judges: unanimity of the other three passes 8 of 54. One synthesis call lands between the vote and three rounds of debate (0.849 against 0.870) and recovers one of ten all-wrong problems with an answer none of the members had given; on the saturated pool it recovered none of sixteen.

**Routing we did not measure; by the published tables it collects a minority of its pool's gap and never exceeds the pool's oracle.** On our pool a per-problem-set route collects 0.7 of at most 2.4 points; Trinity collects about 0.15 of its gap and stays under its oracle on every task; the Conductor and Fugu report no oracle, so their share is not computable.

**Decomposition adds nothing on problems a model cannot solve whole, and lifts a model that runs out of tokens.** On 14 hard numeric problems split pass@4 is 0.21 against 0.21 (0 of 7 never-solved problems solved in eight tries each); on 38 problems where one pass is cut off by the cap it gains +0.34 [+0.18, +0.50], above the model's own pass@16, a gain consistent with budget although the split attempt used fewer tokens and the mechanism was not isolated; on medium and easy problems it costs DeepSeek 41 points; handing the parts to other models is never better than splitting within one beyond noise. At equal money the single model ties or wins on every DeepSeek stratum but one, while GLM's split attempt, which uses fewer tokens, is better at both spends.

## 3.4 What explains the pattern: one difficulty factor

Three independent methods say that what the members share is the difficulty of the item and almost nothing else. The one-factor Gaussian copula at the tetrachoric correlation reproduces the joint-failure rate and the majority within one to three points on the SWE-bench top seven (0.107 against 0.110; 0.788 against 0.794) and in every accuracy band of three public matrices, where the independent Condorcet model over-predicts the majority by 14 and 27 points in the SWE-bench bands above 0.5 (and under-predicts it below 0.5, where its majority goes to zero) and by 1 to 9 points on the two leaderboard matrices . Exactly one eigenvalue of the correctness-correlation matrix lies above the Marchenko–Pastur noise edge [12] on every text set and on the SWE-bench top 7 and top 30 . After a fitted two-parameter item-response model the residual correlation among the SWE-bench top seven is +0.05 (residual $n_{\text{eff}}$ 5.3 against 1.4 raw) and within every lower rank window −0.009 to +0.009. Inside an accuracy band members are more correlated than in the full pool, not less ($\bar{\varphi}$ 0.35 against 0.30; 0.47 against 0.25): members of equal accuracy fail on the same items. What remains on text is a small within-vendor excess, +0.06 to +0.11 in residual correlation, 8 to 14 per cent of residual variance. On agents the same reading holds for reruns and for safety: one agent rerun four times is worth 1.4 to 1.9 independent runs, across-model correlation on τ-bench (0.29–0.47) is below within-model, and on OS-Harm misuse breaches are shared across vendors (ρ 0.67) while injection breaches are only weakly shared (ρ 0.28), with no measurable difference between how breaches and task errors co-occur (Section 11). This is why rearranging operators cannot gain: the vote cannot condition on the axis that produces the shared failures. And it is why the label audit matters: of the 48 text problems on which all seven models were scored wrong, 45 were defects of the benchmark; joint-failure rates reported without such an audit measure the benchmark as much as the models.

## 3.5 What was known, and what is new

None of the ingredients is unprecedented. That frontier models fail on the same items and that a panel of them carries about two effective votes has been reported from leaderboard matrices (Chen, [15]; Bertalanič and Fortuna, [16]). That diversity scores predict the gain of a vote poorly was shown by Kim [14], and our earlier replication (Section 12 [17].3) traced the failure to the linear control rather than to the absence of a signal. That debate raises agreement, that orchestrators beat the best model in their pools by one to nine points on their own tables, and that a single agent at equal budget matches a team on coding tasks, are all in the literature (Section 12).

The contributions of Section 1.5 follow from this ledger; in one line each: (1) one yardstick, the share of the best-to-oracle prize collected, applied to seven operators on one audited corpus and four public data sets, with the best member cross-validated; (2) the finding that the null for voting is not a ceiling effect, since it holds

with 1 to 9 points of room in non-frontier pools and 14 to 32 points in mid-ranked bands, where the independent Condorcet model fails by up to 27 points; (3) three routes to the one-difficulty-factor reading, on text and on agents; (4) the label audit as a result; (5) debate taken apart with a self-only control, truncation accounting, and a pre-registered run on a pool with unshared errors whose four predictions all missed; (6) the equal-budget algebra with a fitted draw correlation, giving a break-even rule in points and price ratio; (7) pre-registered co-failure measurements on agents with tools and on safety breaches from released data. We claim no new operator and no theorem; the contribution is an evaluation with the controls the usual accounting lacks, and it suffices for an evaluation venue.

# 4. Data and protocol

## 4.1 The data sets

**Table 4.1. Data sets, their size, who produced the runs, and what each can show. The non-frontier pool (four of the seven text-corpus models) and the two public leaderboard matrices are listed after the text corpus.**

| Data set | Members × problems | Origin | Operators seen on it | Room above best member |
|---|---|---|---|---|
| Text corpus | 7 models (5 vendors) × 738 problems (733 answered by all seven: MMLU-Pro 485 [3], GPQA Diamond 91 [4], GSM-Hard 128 [5], olympiad 29 [6, 7]); plus 8–16 draws per problem for three models on ~60 problems per set; 4 personas × 4 draws; 3-round debates on 225 problems; 5 judges on contested problems | Own runs (temperature 0 for the matrix, 1 for draws); four models through an API with an 8,192-token cap | Voting, self-consistency, debate, selection, certification, personas, decomposition (Section 9), routing arithmetic | ≤ 2.3 points (733 problems); 2.4 on all 738 rows |
| Non-frontier pool (text corpus without Opus, Sonnet, Qwen) | 4 models × the same 733 problems; a pre-registered debate (3 rounds), synthesis and 3-judge run on the 146 problems the four disagree on, 40 all-right controls and 10 all-wrong problems | Own runs (same prompts, caps and grader as the corpus); $2.86 billed plus subscription usage | Voting, debate, synthesis, selection, certification with room above the best member | 1–9 points (voting); 3.6 (debate run) |
| Public correctness matrices | Chen MMLU-Pro 52 [15, 18] × 200, MATH-Hard 67 × 298, MATH-500 52 × 200; Open LLM Leaderboard [8] MMLU 28 × 14,042 | Released by other authors | Voting in matched-accuracy bands, held-out best member | 14–25 points in the 0.5–0.7 band |
| Public debate traces | 3 agents × 300 or 200 problems × 16 configurations (9,600 traces) | Released by other authors | Debate | Varies; weak models have room |
| SWE-bench Verified leaderboard [1, 1] | 175 submissions × 500 bugs (resolved / not) | Leaderboard, results files only | Voting, oracle, co-failure structure | 9.8 points (top 7) |
| τ-bench trial logs [9, 10] | One agent × 4 (or 8) trials × tasks, several models | Released with the benchmark | Self-consistency of an agent, oracle | 14–25 points (any-of-four) |
| OS-Harm logs [2] | 5 agents (4 vendors) × 150 tasks (misuse 50, injection 50, misbehavior 50) | Released by the benchmark's authors | Co-failure of safety breaches | not an accuracy setting |

| | | | | |
|---|---|---|---|---|
| Cost-matched draws | 1 model × 16 draws × ~60 problems per set, scored at every token budget; three comparator models' single answers with prices | Own runs, re-scored offline | Voting at equal budget, operators at own cost, dollars | Set-dependent |

The text corpus is the only place where all operators meet on identical problems; it is also saturated. The non-frontier pool keeps the problems and the operators and removes the ceiling for the four weaker members. The agent data sets and the public matrices have room above the best member but allow only the operators that an outcome matrix supports: voting, the oracle, and the structure of co-failure. Keeping all three kinds is deliberate: the first shows what operators do when the ceiling is close, the others what they do when it is not.

## 4.2 Scoring and the label audit

Answers on the text corpus are graded by exact or normalised match to the gold label (expressions on the olympiad set are normalised for fractions, roots and formatting). After the first analysis, every problem on which all seven models, or all six debate agents, were scored wrong was re-solved by a strong model and checked by hand. Of 48 such problems, 21 had a wrong gold label and were relabelled, 22 were ambiguous or ill-posed and were excluded, and 3 stand with every model wrong (plus two on which the agents' answers were defensible readings) . Every stored row was then re-graded. The audit is *one-sided*: problems on which some model agreed with a wrong gold label were not re-read, so the post-audit joint failure of 1 in 733 is a lower bound, and the relabelled items favour the strongest model slightly (it is right on 9 of 9 relabelled MMLU-Pro items, the others on 7 or 8).

## 4.3 Truncation

Four of the seven models ran through a provider with an 8,192-token completion cap, and two of them reason at length; 229 of the matrix problems have at least one answer cut off by the cap . A truncated answer is reported as *missing* in the headline tables (the problem is dropped from "problems all seven finished") and as *wrong* in a secondary table. The distinction matters most for debate (Section 6), where almost all of the measured gain lies on problems that an agent had not finished at round 0, and for self-consistency on the olympiad set.

## 4.4 Statistics

Differences between an operator and a member on the same problems are tested with McNemar's exact test on the discordant counts (problems the operator gets right and the member wrong, and the reverse); these counts are reported next to every such comparison. Intervals are percentile bootstraps over problems (and over members where the panel is resampled). "Best member" comparisons are given three ways where the data allow: in sample, split-half (best chosen on one random half, scored on the other, 2,000 splits) and with the member fixed in advance. Correlations are Pearson $\varphi$ on 0/1 vectors; tetrachoric values use the cos-π approximation and are flagged as upward-biased when accuracies exceed 0.9. The item-response fits are two-parameter logistic models estimated by marginal maximum likelihood with the problems with no correct answer dropped (on SWE-bench, 474 of 500). No correction for multiple comparisons is applied; the number of comparisons per section is stated. The comparisons that reach $p < 0.05$ uncorrected are few: the in-sample best over the vote on MMLU-Pro ($p = 0.049$), the vote over a fixed mid-ranked member ($p \le 0.008$), debate under a fixed tie order ($p = 0.001$) and the non-frontier debate on contested problems ($p = 0.031$).

## 4.5 Pre-registration

The following were pre-registered, with the hypothesis, the quantity and the decision rule committed before the data were read: the SWE-bench co-failure study (four predictions, Section 11), the τ-bench study, the OS-

Harm study, the decomposition study (two protocol revisions, both recorded), the cost-matched study, the item-response residual studies, and the replications of Kim et al. The non-frontier debate, synthesis and judge run (Sections 6.4, 7.5) was pre-registered with four predictions, all of which failed, each toward a larger effect than predicted. The text-corpus matrix, debate, judge and persona studies were exploratory and were revised three times after the label audit, a truncation audit and a reading of the code against the text; in those sections every tie rule, stopping rule and "strongest agent" that was chosen after seeing the data is marked as such. Of the four SWE-bench predictions three were missed, each toward more correlation than expected ($\varphi$ predicted 0.3–0.6, observed 0.68; oracle predicted at least 10 points above the best, observed 9.8; all-wrong predicted under 5 per cent, observed 5.2) and one held (majority at or below the best); Section 11.2 lists them.

### 4.6 Reproducibility

The non-frontier-pool analysis of Section 5.4, the below-ceiling analysis of Section 5.5 and the stored-log part of Sections 6.4 and 7.5 are zero-call recomputations (scripts weak-pool-voting.py, below-ceiling-swebench.py, below-ceiling-chen.py, weak-pool-operators-phase1.py, kept with the draft); the four-model debate, synthesis and judge run is weak-pool-operators-phase2.py with its pre-registration file and per-call records.

Code and data for the text-corpus analyses, the public-trace debate study, the diversity-control replication and the decomposition study are released with persistent identifiers (five companion repositories with software DOIs [19, 20, 21, 22, 23]). The SWE-bench, τ-bench and OS-Harm matrices are derived from released files with scripts that are prepared but, as of this draft, not yet released; the cost-matched study is drafted but unpublished. Where a number in this paper depends on an unreleased script, the section says so.

## 5. Plurality voting, self-consistency and the sources of diversity

**What this section shows.** Plurality voting collects nothing over a best member chosen on held-out problems; what it gives is insurance against a member fixed in advance that turns out to be mid-ranked. We test it on three stands with growing prizes (oracle minus best member): the text corpus of 733 problems and seven models, prize 0.0–2.3 points (Section 5.3); pools without the frontier models, 1.1–4.3 points for four models and 1.6–8.8 for three (Section 5.4); and members of mid-ranked accuracy on three public correctness matrices, 14 to 32 points (Section 5.5). On none does the vote collect the prize. Section 5.6 gives the reason: the members fail on the same hard problems, and one difficulty factor accounts for the vote on every stand.

### 5.1 The operator

**Plurality voting** (majority voting) asks every member for one answer and returns the answer given most often; answers count as the same when they are the same option letter or the same normalised number or expression, and ties go to the first member in a fixed order. **Self-consistency** is the same operator applied to repeated draws of a single member at non-zero temperature. Both are selection operators in the sense of Section 2: they can only return an answer some member produced, so neither can exceed the **oracle** (the share of problems at least one member answers correctly).

Two formulas frame what voting can add. Under the **independent Condorcet jury model [24]**, n voters each correct with probability p independently give P(majority correct) = $\Sigma_{k > n/2}\, C(n,k)\, p^k\, (1-p)^{n-k}$, which tends to 1 as n grows whenever $p > 1/2$. Correlated errors break this. The **effective number of independent votes** (Kish), $n_{eff} = n / (1 + (n-1)\cdot\bar{\varphi})$, where $\bar{\varphi}$ is the mean pairwise correlation of correctness, is the number of independent voters whose majority would have the same variance.

From the literature: self-consistency over sampled reasoning paths was introduced by [25]. [15] argues, over pools of up to 67 models, that the binding quantity for any combination is the all-wrong rate rather than mean

pairwise correlation, since every selection policy is capped at the oracle. [14] audits diversity metrics over 31,900 subsets of 30 models and finds that much of what they predict about majority-vote gain is member capability re-expressed (Section 12).

## 5.2 Data and protocol

**Text corpus** (Section 4): one temperature-0 answer per problem from seven models in five families (Claude Haiku, Opus and Sonnet; DeepSeek V3.2; Mistral Large; Qwen3.7-plus; GLM-5.3-flash). The voting analysis uses the 733 problems answered by all seven: MMLU-Pro 485 [3], GPQA Diamond 91 [4], GSM-Hard 128 [5], olympiad 29 [6, 7]. Four models ran under an 8,192-token cap, and problems where a capped model was cut off are excluded: 106 of 197 GPQA Diamond and 61 of 90 olympiad problems. The analysis was not pre-registered; split-half selection and Opus as a fixed comparator were chosen after the in-sample results, after the in-sample results had been seen.

**Pools without the frontier models**: the same code and 733 problems with Opus, Sonnet and Qwen removed, giving four models (Haiku, DeepSeek V3.2, Mistral Large, GLM-5.3-flash), the three without GLM (an odd pool), and the four plus Qwen as a control. The cut was decided after Table 5.1 was seen; "non-frontier" is a label, not a ranking (GLM leads the four on MMLU-Pro and GPQA Diamond).

**Public correctness matrices**: the SWE-bench Verified leaderboard [1, 1] (175 submissions × 500 bugs, Section 11), Chen's MMLU-Pro matrix [15, 18] (52 models × 200 items) and the Open LLM Leaderboard [8] MMLU matrix [8] (28 models × 14,042 items). On each we kept the members whose accuracy falls in a band, cut in sample. On SWE-bench the majority is over the whole band and over 200 random odd panels of seven; on Chen and the Leaderboard it is the held-out majority of all size-3 and size-5 subsets, the best member chosen on a training half and scored on the other (7,021 held-out items on the Leaderboard; protocol of our earlier replication, Section 12 [17].3). Pools and bands are zero-call recomputations decided after Tables 5.1 and 5.2 were known (Section 4.5).

**Draws of one model**: Haiku, DeepSeek V3.2 and GLM-5.3-flash at T = 1, k = 16 draws per problem (k = 8 for Haiku on GPQA Diamond and the olympiad set), about 60 problems per set; for roles, DeepSeek V3.2 under four persona prompts, four draws each.

## 5.3 The saturated pool

**Table 5.1. Plurality voting over seven models on the 733 problems all seven answered. Best (in-sample) is the most accurate member chosen on the same problems it is scored on. Split-half: best chosen on a random half, scored on the other half against the plurality on that half, mean over 2,000 splits with 95% interval; positive favours the best member. Mean φ and $n_{eff}$ over the 21 model pairs (olympiad: φ defined on 10 of 21 pairs).**

| Problem set | n | Best (in-sample) | Plurality | Split-half best – plurality (points) | Oracle | Mean φ | $n_{eff}$ |
|---|---|---|---|---|---|---|---|
| MMLU-Pro | 485 | 0.975 (Opus) | 0.957 | +1.8 [0.0, +3.3] | 0.998 | 0.38 | 2.14 |
| GPQA Diamond | 91 | 0.989 (GLM) | 0.978 | −0.5 [−2.2, 0.0] | 1.000 | 0.34 | 2.29 |
| GSM-Hard | 128 | 0.992 (Sonnet) | 0.984 | −0.3 [−3.1, +1.6] | 1.000 | 0.39 | 2.09 |
| Olympiad | 29 | 1.000 (Sonnet, Qwen) | 1.000 | −5.1 [−6.7, 0.0] | 1.000 | 0.20 | 3.21 |

The best member scores 0.975–1.000 and the oracle 0.998–1.000, so the prize is at most 2.3 points (MMLU-Pro, 484 against 473 of 485). In-sample, the plurality is 0.0 to 1.9 points below the best member. On MMLU-Pro the best member (Opus) is right where the plurality is wrong on 13 problems and the reverse on 4 (exact

McNemar p = 0.049); elsewhere the discordant counts are 1 vs 0, 1 vs 0 and 0 vs 0. That comparison flatters the single model, chosen on the problems it is scored on. With split-half selection the best member leads only on MMLU-Pro (+1.8 points, interval touching zero), and the plurality is level or ahead on the other three sets. Against Opus fixed in advance (the strongest model, uncapped), over the 733 problems, Opus answers 714 and the plurality 708 (13 vs 7 discordant, p = 0.26). **On these saturated sets the vote and the best member are statistically indistinguishable, and the vote collects about none of the prize.** The seven models act like about two independent voters: mean φ 0.34–0.39, $n_{eff}$ 2.09–2.29 on the three larger sets.

*Caveats*. The MMLU-Pro p is uncorrected for four sets and ranges 0.013–0.092 over tie orders. Joint failure is small partly through the label audit (Section 4.2): of 48 problems all models got wrong, 45 were benchmark defects (21 wrong gold labels, 22 ambiguous or ill-posed, 2 grading failures), leaving 1 of 733. The audit was one-sided, so 1 of 733 is a lower bound; with relabels reverted MMLU-Pro has 10 of 485 all-wrong. Of 229 problems with a cut-off answer, every answering model is wrong on 19 (8.3%), mostly unaudited.

## 5.4 Pools without the frontier models

A natural objection to Table 5.1 is that the pool has no room because it contains the strongest models. Table 5.2 removes them.

**Table 5.2. Plurality voting in pools without the frontier models, on the same 733 problems as Table 5.1 (first-in-order tie rule). Best (in-sample) is chosen on the same problems it is scored on; headroom = oracle – in-sample best. Split-half and McNemar as in Table 5.1; the fixed member is DeepSeek V3.2, chosen in advance as a mid-ranked member.**

| Pool | Problem set | n | Best (in-sample) | Plurality | Oracle | Headroom | All wrong | Mean φ | $n_{eff}$ | Split-half best – plurality | DeepSeek vs plurality (won / lost, p) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Four | MMLU-Pro | 485 | 0.938 (GLM) | 0.924 | 0.981 | 4.3 | 9 | 0.43 | 1.74 | +1.4 [−0.8, +3.3] | 7 / 22, 0.008 |
| Four | GPQA Diamond | 91 | 0.989 (GLM) | 0.901 | 1.000 | 1.1 | 0 | 0.28 | 2.19 | +8.9 [+4.3, +15.2] | 2 / 5, 0.45 |
| Four | GSM-Hard | 128 | 0.977 (Haiku) | 0.977 | 0.992 | 1.6 | 1 | 0.38 | 1.88 | −0.6 [−4.7, +1.6] | 1 / 4, 0.38 |
| Four | Olympiad | 29 | 0.966 (Haiku) | 1.000 | 1.000 | 3.4 | 0 | 0.15 | 2.74 | −6.6 [−13.3, +0.0] | 0 / 2, 0.50 |
| Three | MMLU-Pro | 485 | 0.905 (Haiku) | 0.905 | 0.961 | 5.6 | 19 | 0.48 | 1.54 | −0.4 [−3.3, +1.2] | 10 / 16, 0.33 |
| Three | GPQA Diamond | 91 | 0.868 (DeepSeek) | 0.846 | 0.956 | 8.8 | 4 | 0.40 | 1.66 | −0.2 [−6.5, +4.3] | 4 / 2, 0.69 |
| Three | GSM-Hard | 128 | 0.977 (Haiku) | 0.977 | 0.992 | 1.6 | 1 | 0.36 | 1.74 | −0.4 [−6.2, +0.0] | 2 / 5, 0.45 |
| Three | Olympiad | 29 | 0.966 (Haiku) | 0.931 | 1.000 | 3.4 | 0 | 0.20 | 2.14 | +1.5 [−6.7, +6.7] | 1 / 1, 1.00 |

| Pool | Problem set | n | Best (in-sample) | Plurality | Oracle | Headroom | All wrong | Mean φ | $n_{eff}$ | Split-half best – plurality | DeepSeek vs plurality (won / lost, p) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Four + Qwen | MMLU-Pro | 485 | 0.955 (Qwen) | 0.940 | 0.984 | 2.9 | 8 | 0.42 | 1.87 | +1.3 [−1.2, +3.3] | 3 / 26, <0.001 |
| Four + Qwen | GPQA Diamond | 91 | 0.989 (GLM) | 0.934 | 1.000 | 1.1 | 0 | 0.30 | 2.30 | +4.4 [+0.0, +8.7] | 1 / 7, 0.07 |

The weaker pools have room: the prize is 1.1 to 4.3 points for the four and 1.6 to 8.8 for the three, and the problems no member solves rise from 1 to 10 and 24 of 733 (78, or 8.2%, of the three's own 949 finished problems). The plurality collects none of it. In 10 of the 12 pool-and-set rows the in-sample plurality is at or below the in-sample best; in the other two it is above by one problem. The split-half interval contains zero in every row but GPQA Diamond in the four-model pool, where GLM scores 0.989 against 0.76–0.87 for the other three and the equal-weight vote lands 8.9 points [+4.3, +15.2] below it. This is the case the heterogeneous Condorcet model of Section 2.2 predicts: when one member is far more accurate than the rest, the vote cannot exceed it. Adding Qwen halves the gap; removing GLM removes it with the accuracy.

What the vote buys is insurance against a member fixed in advance. Against DeepSeek V3.2 the plurality wins on MMLU-Pro in every pool (7 / 22, $p = 0.008$ for the four; 3 / 26, $p < 0.001$ with Qwen) and over all 733 problems (10 / 33, $p = 0.001$ for the four): the vote of three to five mid-ranked models scores about where the best of them does, without knowing which one that is. Correlation does not fall as the pool weakens: mean $\varphi$ is 0.28 to 0.48 on the three larger sets in every pool, 0.44 to 0.58 on MMLU-Pro and GSM-Hard over the weak pools' own finished problems, and $n_{eff}$ stays at 1.4 to 3.1. A pool with headroom is not a pool with more independent voters.

*Caveats*. Outside MMLU-Pro each set is one or two problems wide, with intervals as wide as [−13.3, +0.0]; the one-sided audit biases against the weak pools; twelve McNemar tests are uncorrected.

## 5.5 Mid-ranked members of public matrices

The reference point is the top of the SWE-bench leaderboard (Section 11.2): for the seven best submissions the prize is 9.8 points (best 0.792, oracle 0.890) and their majority resolves 0.794 (12 bugs won, 11 lost), a share of 0.02, with φ 0.68 and $n_{eff}$ 1.38; with one submission per model the majority is 0.784, below 0.792. Table 5.3 moves down to members of equal, mid-ranked accuracy.

**Table 5.3. Plurality voting among members of mid-ranked accuracy. On SWE-bench, best is in-sample (chosen on the same 500 scored on) and split-half is as in Table 5.1; on Chen and Leaderboard, best is held out (chosen on a training half, scored on the ot Headroom = oracle – best. "Panels beat best" = share of random odd panels of seven (SWE-bench) or of held-out subsets (Che Leaderboard) whose majority exceeds their own best member. Independent = heterogeneous Condorcet majority from the mem accuracies with no dependence; copula = one-factor model at the band's tetrachoric ρ. Residual r = 2PL residual correlation wit band (null −0.006). Bands are cut in sample.**

| Matrix, band | n members | Best | Majority | Oracle | Headroom | Mean φ | $n_{eff}$ | Split-half best – majority | Panels beat best | Independent majority | Copula majority | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SWE-bench, acc 0.6–0.7 | 31 | 0.696 | 0.680 | 0.884 | 18.8 | 0.61 | 1.61 | −0.2 [−3.6, +2.8] | 0.17 | 0.953 | 0.655 | |

| Matrix, band | n members | Best | Majority | Oracle | Headroom | Mean φ | $n_{eff}$ | Split-half best − majority | Panels beat best | Independent majority | Copula majority | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SWE-bench, acc 0.5–0.6 | 31 | 0.598 | 0.564 | 0.854 | 25.6 | 0.56 | 1.75 | +1.2 [−3.2, +4.8] | 0.14 | 0.700 | 0.550 | |
| SWE-bench, acc 0.4–0.5 | 28 | 0.492 | 0.448 | 0.814 | 32.2 | 0.53 | 1.82 | +2.5 [−2.0, +5.6] | 0.01 | 0.206 | 0.423 | |
| SWE-bench, ranks 81–140 | 60 | 0.584 | 0.494 | 0.862 | 27.8 | 0.53 | 1.86 | +6.7 [+2.0, +10.0] | 0.00 | 0.375 | 0.481 | |
| Chen MMLU-Pro, acc 0.5–0.7, subsets of 3 | 15 | 0.648 (held out) | 0.645 | 0.847 | 19.9 | 0.35 | 1.4–1.8 | +0.3 [−0.9, +1.9] | 0.45 | best +1.1 | best −2.3 | |
| Chen MMLU-Pro, acc 0.5–0.7, subsets of 5 | 15 | 0.653 (held out) | 0.655 | 0.903 | 25.0 | 0.35 | 1.5–2.1 | −0.2 [−2.3, +2.2] | 0.49 | best +3.6 | best −2.9 | |
| Leaderboard MMLU, acc 0.5–0.7, subsets of 3 | 14 | held out | best −4.1 | – | 20 | 0.47 | 1.4–1.8 | +4.1 | 0.007 | best +0.7 | best −3.3 | |
| Leaderboard MMLU, acc 0.5–0.7, subsets of 5 | 14 | held out | best −5.6 | – | 14 | 0.47 | 1.5–2.1 | +5.6 | 0.000 | best +3.0 | best −4.4 | |

In the SWE-bench bands 0.6–0.7, 0.5–0.6 and 0.4–0.5 (31, 31 and 28 submissions) the prize is 18.8, 25.6 and 32.2 points, with 58, 73 and 93 of 500 bugs no member resolves; the majority resolves 0.680, 0.564 and 0.448 against a best of 0.696, 0.598 and 0.492. Out of sample the best member leads by −0.2 [−3.6, +2.8], +1.2 [−3.2, +4.8] and +2.5 [−2.0, +5.6] points, and 17%, 14% and 1% of random odd panels of seven beat their own best. Where a window is wide in accuracy (ranks 81–140, 0.39 to 0.58) the weak members pull the vote down and the best leads by +6.7 [+2.0, +10.0], GPQA Diamond of Table 5.2 at scale. On Chen's MMLU-Pro band (15 models; prize 19.9 points for subsets of three, 25.0 for five) the held-out majority is −0.3 [−1.9, +0.9] and +0.2 [−2.2, +2.3] points from the held-out best and beats it in 45–49% of subsets, a coin flip. On the Leaderboard band (14 models, 7,021 held-out items, prize 20 and 14) it is 4.1 and 5.6 points below and beats the best in 0.7% and 0.0% of subsets. The matched-accuracy band is the best case for voting, and the best case is a tie with 20 points on the table.

The 82 SWE-bench submissions that declare an internal best-of-*n* selector match the others within every band on resolved rate (within 0.03), on φ and on the union-minus-best gap; what a selector with an external verifier recovers of its own candidates' union is not measured (Section 7.6).

*Caveats*. Bands are cut on the instances they are scored on; leaderboard submissions are not independent agents (Section 11.2); the selector flag is self-declared and resolved rates are post-selection; Chen's MATH bands (four and five members) are not reported; sixteen window tests and thirty subset rows are uncorrected.

## 5.6 What explains the vote: one difficulty factor

Tables 5.1–5.3 have one explanation: the problems the best member misses are the ones the others miss too.

**Independence over-predicts the vote; one factor predicts it.** From the members' accuracies alone, the heterogeneous Condorcet model over-predicts the majority by 14 and 27 points in the two SWE-bench bands above 0.5, and under-predicts it in the 0.4–0.5 band where its majority goes to zero (0.953 against 0.680 in 0.6–0.7) and predicts all-wrong rates below 0.0001 where 0.09 to 0.19 are observed. On Chen it predicts +1.1 and +3.6 points over the held-out best, positive for 67–85% of subsets; on the Leaderboard +0.7 and +3.0: misses of 1 to 9 points. The one-factor Gaussian copula of Section 2 (member *i* correct iff $(\sqrt{\rho})\cdot D + (\sqrt{(1-\rho)})\cdot E_i$ exceeds a threshold set by its accuracy), at the stand's tetrachoric ρ (0.51 to 0.91), predicts the all-wrong rate and the majority within 1 to 3 points everywhere: −2.3 and −2.9 points against the held-out best on Chen, −3.3 and −4.4 on the Leaderboard, and at the SWE-bench top seven (ρ 0.91) an all-wrong rate of 0.107 against 0.110 and a majority of 0.788 against 0.794. On the text corpus mean tetrachoric ρ is 0.83, 0.74 and 0.86 on MMLU-Pro, GPQA Diamond and GSM-Hard, far above φ. There the copula's majority (counted on correctness, more than half of the seven right, which can fall below the plurality of Table 5.1) is 0.941, 0.957, 0.975 and 0.989 against 0.946, 0.967, 0.984 and 1.000 observed; it over-predicts joint failure on MMLU-Pro (0.011 against 0.002), which matches at ρ = 0.60 rather than 0.83. One difficulty axis is enough for the vote and, if anything, overstates how often all seven fail together.

**Matched accuracy does not decorrelate.** Inside a band $\bar{\varphi}$ is higher than in the full pool: 0.35 against 0.30 on Chen, 0.47 against 0.25 on the Leaderboard; on SWE-bench it falls only from 0.68 to 0.53 down the ranking. A band trio carries 1.4 to 1.8 effective votes, a quintet 1.5 to 2.1.

**Nothing shared remains after a fitted difficulty axis.** The 2PL residual correlation within every SWE-bench window below the top seven is −0.009 to +0.009 (null −0.006). Among the top seven, in a pre-registered analysis (Section 4.5), it is +0.05, turning the raw $n_{eff}$ of 1.4 into 5.3 on residuals: net of difficulty the seven best are near independent, but the vote acts on raw answers.

**Families add a small excess.** On Chen's three matrices and an ARC matrix, same-family models are +0.06 to +0.11 more residually correlated than cross-family pairs, and family explains 8–14% of residual variance (pre-registered; larger than predicted on Chen). It is second-order next to difficulty.

So the vote rearranges answers whose failures are already shared: it ties the best member at equal accuracy and loses where one member dominates or the band is wide. *Caveats.* The cos-π tetrachoric is biased upward at the text corpus's accuracies; [15] reports the copula under-predicting joint failure under a common-shock model, which we do not see on a saturated, audited corpus; the residual analyses are on public matrices, not our seven models.

## 5.7 Self-consistency

Draws of one model are more correlated than different models: for Haiku and DeepSeek within-model φ is 0.45–0.75 and $n_{eff}$ 1.3–2.1 for 8 or 16 draws; GLM's draws are mostly identical near ceiling (φ 1.00 on MMLU-Pro and GPQA Diamond, 26–55 problems).

Against the *mean* single draw (not the best draw), the majority of k draws gains −0.6 to +1.7 points on MMLU-Pro and GSM-Hard, and on GPQA Diamond +4.3 for Haiku (0.740 → 0.783, 60 problems, k = 8) and +6.7 for DeepSeek (0.806 → 0.873, 55 problems, k = 16). The largest figure is DeepSeek on the olympiad set, 0.753 → 0.898, +14.5 points, on the 49 of 60 problems where all sixteen draws finished; on those 49 it is +12.2 over the temperature-0 answer (0.776) and +6.1 (about three problems) over the first draw (0.837), the top of the range 0.633–0.837 across draw positions. On all 60, with cut-off draws as missing votes, the majority scores 0.833; the mean draw on all 60 was not computed.

Kish's $n_{eff}$ does not predict these gains: for $1 \le n_{eff} \le 2$ the binomial majority of $n_{eff}$ independent votes equals the mean draw, predicting nothing, yet the observed majority exceeds it on 6 of the 8 Haiku and DeepSeek rows, by up to 13.8 points (DeepSeek olympiad: 0.760 predicted, 0.898 observed). It summarises overlap; it does not predict the vote. Whether the gains survive once the draws' tokens are charged is Section 10's question. *Caveats*. No gain here has an interval; each is a few problems of 49–60, uncorrected.

### 5.8 Sources of diversity: family, role, temperature

**Family.** Over the 35 triples of the seven models the plurality is on average 1.4, 3.5, 0.1 and 0.8 points below the triple's in-sample best, the best triple at +0.2, 0.0, +1.6 and +3.4; the family excess of Section 5.6 is too small to change this.

**Role.** Between DeepSeek draws under different personas minus between plain draws, φ differs by +0.17 [−0.39, +0.31] on MMLU-Pro (59 problems), −0.03 [−0.22, +0.16] on GSM-Hard (56) and +0.07 [−0.08, +0.23] on the olympiad set (13), bootstrap over problems, 2,000 resamples; GPQA Diamond was not run. Every interval includes zero, and two point estimates lean toward personas being *more* alike. Four personas give $n_{eff}$ 1.16–1.51; their majority scores 0.945, 0.902 and 0.769 against 0.932, 0.911 and 0.769 for four plain draws.

**Temperature.** Draws of one model are more alike than different models (Section 5.7). *Caveats*. On 13–59 problems a persona prompt is indistinguishable from plain resampling; intervals up to ±0.35 in φ do not exclude a modest effect either way.

### 5.9 What this shows and what it does not

Plurality voting collects nothing over a held-out best member and gives insurance against a fixed mid-ranked one, because the members share one difficulty factor (Section 5.6).

- **Text corpus, seven models.** Prize 0.0–2.3 points; the vote is level with the best member out of sample and with Opus fixed in advance. Not shown: low joint failure beyond problems all seven finished, after a one-sided audit (Sections 5.2, 13).
- **Pools without the frontier models.** Prize 1.1–4.3 (four) and 1.6–8.8 (three); the vote collects none, loses where one member dominates, and beats DeepSeek V3.2 fixed in advance. Not shown: which mid-ranked member to pick; pools chosen after Table 5.1.
- **Mid-ranked members of public matrices.** Prize 14 to 32 points; the majority ties the held-out best on Chen, trails it on SWE-bench and the Leaderboard, and collects 0.02 of the 9.8 points at the SWE-bench top. Not shown: a selector with an external verifier (Section 7.7) or agents run as one ensemble (Section 11).
- **Self-consistency and diversity.** A few points on GPQA Diamond and the olympiad set, without intervals and before tokens are charged (Section 10); personas and families buy no independent votes.

## 6. Debate: multi-round exchange as an operator

**What this section shows.** On a saturated pool, where the best member already sits within a point or two of the oracle, debate collects nothing. Its gain over voting there is cap recovery (answers the token cap cut off reappear), re-elicitation (in the six-copy control about half the gain appears when a member is simply asked again, with no peers) and, on complete problems, a single fix. In a pool of four non-frontier models whose errors are not shared, a pre-registered debate collects about 1.3 of the 3.6 points between the best member and the oracle, so one to two points of a roughly four-point prize. In every setting debate loses coverage (problems

some member had solved end with every member wrong) and the panel collapses toward a single effective voter.

## 6.1 The operator and its formal statement

In **debate**, each of $n$ members first answers a problem on its own (round 0). In every later round each member sees the current answers of all the others and writes a revised answer. After $R$ rounds the final answers are combined by plurality voting. Debate is therefore not sampling: self-consistency redraws from a fixed per-problem answer distribution, while debate shifts that distribution in response to the peers.

Formally, let $a_{i,r}(x)$ be member $i$'s answer to problem $x$ in round $r$. Round 0 is $a_{i,0} \sim P_i(\cdot \mid x)$. For $r \geq 1$, $a_{i,r} \sim P_i(\cdot \mid x, a_{1,r-1}, \ldots, a_{n,r-1})$. The output is the plurality of $\{a_{i,R}\}$. The control is the round-0 plurality, which is plain voting over the same members, and the gain is the round-$R$ plurality minus the round-0 plurality. Across rounds we track the mean pairwise correlation $\bar{\varphi}$ and the effective number of independent votes $n_{\text{eff}} = n \,/\, (1 + (n-1)\cdot\bar{\varphi})$ (Section 2). The ceiling is the oracle over the members, and debate can lower that ceiling itself, because a member that was right can be argued out of its answer. *Synthesis* (mixture-of-agents) is the one-call relative of debate: a single member reads several answers and writes one.

## 6.2 What the literature claims

One test-time scaling theory ([26]) shows that voting over exchangeable samples converges to the *modal-hit rate*, the share of problems on which the most common answer is correct. TUMIX ([27]) reports that over-refinement collapses diversity while coverage stays above accuracy. Work on sycophancy ([28]) finds that debate does not consistently beat the best single model. A mean-field analysis ([29]) predicts that conformity turns a small shared bias into a biased consensus. A compute-matched study on MMLU-Pro ([30]) ranks mixture-of-agents above debate, and debate above self-consistency, with most gaps inside ±3 points. Several expect debate among copies to converge on the model's preferred answer; we test this against the *MAP ceiling*, the share of problems on which the model's modal answer is correct.

## 6.3 The saturated pool: six mixed models and six copies of one

**Protocol.** Six models from different families (DeepSeek-V3.2, GLM-5.3-Flash, Mistral-Large, Qwen3.7-Plus, Claude Haiku, Claude Sonnet) debated for three synchronous rounds on 225 problems of the 733-problem text corpus, 54–58 from each of MMLU-Pro, GPQA Diamond, GSM-Hard and olympiad mathematics. Round 0 is each model's temperature-0 answer (Section 4). An answer cut off by the token cap counts as a missing vote, not a wrong one: there were 86 truncated rows at round 0 and 21 at round 3. The controls are six DeepSeek copies, whose round 0 is six temperature-1 draws. They run in three arms: plain debate; a *self-only* arm, in which each copy revises its own answer without seeing peers, which measures *re-elicitation*; and a *defer* arm, in which each copy is told to follow the majority.

**Table 6.1. Debate on the saturated pool (three rounds; a cut-off answer is a missing vote). "Fixed / broken" counts the problems where the round-3 majority is right and the round-0 majority wrong, and the reverse. The copy arms are shown on the problems each arm completed; the gains in the text use the 230 problems that all three arms completed.**

| arm | problems | round-0 majority | round-3 majority | fixed / broken | mean ϕ round 0 → 3 |
|---|---|---|---|---|---|
| Mixed six-model panel, fixed tie order | 225 | 0.924 | 0.973 | 11 / 0 | 0.34 → 0.85 |
| Mixed panel, ties split evenly | 225 | 0.934 | 0.973 | – | – |

| arm | problems | round-0 majority | round-3 majority | fixed / broken | mean φ round 0 → 3 |
|---|---|---|---|---|---|
| Mixed panel, strongest agent breaks ties | 225 | 0.951 | 0.973 | 5 / 0 | – |
| Six DeepSeek copies, plain debate | 236 | 0.856 | 0.894 | 12 / 3 | 0.58 → 0.97 |
| Six DeepSeek copies, self-only (no peers) | 230 | 0.865 | 0.883 | 8 / 4 | 0.57 → 0.58 |
| Six DeepSeek copies, told to defer to majority | 236 | 0.856 | 0.903 | 13 / 2 | 0.58 → 0.95 |

**Mixed panel.** Under the fixed tie order the majority rises from 0.924 to 0.973, with 11 problems fixed and none broken (exact McNemar p = 0.001). Seven of the 11 fixes are round-0 ties that the fixed order had resolved against the correct answer. When the strongest agent breaks ties, the gain shrinks to 0.951 → 0.973, with 5 fixed, 0 broken and p = 0.06. That agent is chosen in-sample, by round-0 accuracy on these same problems. With ties split evenly the gain is 0.934 → 0.973. In 10 of the 11 fixes at least one agent had been cut off at round 0. On 6 of those 10, the winning majority is made of agents that had already answered, and the cut-off agent simply casts no vote. On the 162 problems all six agents answered at round 0, the majority moves from 0.975 to 0.981, a single fix. We also compare with the best member on each benchmark, chosen in-sample and scored on the problems it answered. On those problems the round-3 majority is +2.0 points above it on GPQA Diamond (49 of 56 answered), level on olympiad, and −1.8 on MMLU-Pro and GSM-Hard.

**Collapse and coverage.** Unanimity rises from 152 to 219 of 225 problems, and mean φ from 0.34 to 0.85, so $n_{\text{eff}}$ falls from 2.21 to 1.15. The number of problems on which the panel is unanimously wrong rises from 0 to 4. A correct agent facing five disagreeing peers switched to a wrong answer in 5 of 6 such cases.

**Copies and re-elicitation.** On the 230 problems, plain debate among the copies gains +3.5 points (0.865 → 0.900, 11 fixed / 3 broken). The defer arm gains +4.3. The self-only arm still gains +1.7 (8 / 4) and leaves φ unchanged (0.57 → 0.58). About half of the copies' gain therefore needs no peers (a point estimate). What peers add is agreement. In plain debate φ rises from 0.58 to 0.97, and the number of problems on which all six copies are unanimously wrong rises from 6 to 22. The mixed panel has no self-only arm, so its first-round jump mixes four things: peer influence, re-elicitation, a new prompt, and fewer cut-off rows.

**Stopping and synthesis.** Suppose each problem stops at its first unanimous round, or at round 3 if there is none. This gives 0.973, the same as three fixed rounds, at 0.42 rounds per problem on average, about a sixth of the debate tokens. The rule was chosen after looking at these data and is scored on the problems it was chosen on. It loses nothing only because no problem was unanimously wrong at round 0. Synthesis by one DeepSeek call recovers 0 of the 16 problems on which none of DeepSeek's draws was correct.

## 6.4 The non-frontier pool, pre-registered

The pool of Section 6.3 leaves almost no room above its best member. To test debate where there is room, we ran a new debate among the four non-frontier models defined in Section 5.4: Haiku, DeepSeek V3.2, Mistral Large and GLM-5.3-flash. The protocol was three synchronous rounds, with the stored answers as round 0 and the paper's prompts, caps, temperatures and grader. It ran on three sets of problems from the 733: 146 *contested* problems, on which at least one of the four is right and one is wrong (MMLU-Pro 95, GPQA Diamond 29, GSM-Hard 16, olympiad 6); the 10 problems all four get wrong; and a random control of 40 problems all four get right. No agent was cut off at round 0 on any of these problems, so cap recovery is absent by construction. Four predictions and their decision rules were committed before the first call. The run billed $2.86 plus about $14 of subscription usage for Haiku (Section 10 compares operators at equal budget). In the

same run one DeepSeek call synthesised an answer from the four round-0 answers, and three of the four models acted as judges (Section 7.5).

**Table 6.2. Pre-registered predictions for the four-model debate, and what happened. "Strongest first" breaks ties toward GLM, the most accurate member on these problems (chosen in-sample).**

| Prediction | Rule | Observed | Held? |
|---|---|---|---|
| Debate gain small | round-3 – round-0 plurality ≤ +3 points on contested problems, strongest first | +6.8 (0.801 → 0.870; 14 fixed / 4 broken, $p$ = 0.031); +15.1 under the fixed order | no |
| Panel collapses | mean ϕ within the four ≥ 0.80 at round 3 | 0.58 (from −0.10 at round 0; $n_{eff}$ 5.7 → 1.5) | no |
| Synthesis creates nothing | 0 of the 10 all-wrong problems correct | 1 of 10, with an answer none of the four had given | no |
| Judges within the vote | panel – plurality within ±3 points, strongest first | +5.5 (0.856 vs 0.801; 17 / 9, $p$ = 0.17) | no |

**Debate.** The predictions assumed this pool would behave like the saturated one; all four missed, all in the direction of a larger effect. On the contested problems the four members' errors are uncorrelated at round 0: $\bar{\varphi}$ is −0.10, and $n_{\text{eff}}$ is above the panel size. There is something to exchange, and three rounds raise $\bar{\varphi}$ only to 0.58. The plurality moves from 0.801 to 0.870 under the strongest-first rule. Under the fixed order the gain is +15.1, so the tie rule alone shifts the round-0 baseline by several points. To project the result onto the 733 problems, we keep the 577 all-right problems that were not debated (the control broke 0 of 40). The plurality then moves from 0.947 to 0.963. The in-sample best member, GLM, scores 0.950 (16 problems won / 6 lost, p = 0.052), and the oracle is 0.986. Debate thus collects about 1.3 of the 3.6 points between the best member and the oracle. It cannot be ranked above that member with confidence, even under the in-sample tie rule most favourable to the vote.

**Coverage.** Eleven contested problems end with every member wrong, nine of them unanimously, so the oracle over the four falls from 1.000 to 0.925 on the contested set. A correct agent facing three disagreeing peers switches to a wrong answer 46 per cent of the time (29 per cent in the six-agent panel). The 10 all-wrong problems gain 2, and the control loses none of 40.

**Synthesis.** At one call per problem, synthesis reaches 0.849 on the 146 contested problems (debate 0.870). It also recovers 1 of the 10 all-wrong problems, with an answer none of the four inputs had given. The 0 of 16 in Section 6.3 therefore describes a saturated pool, not the operator.

**What the stored rows cannot measure.** The stored six-agent debate cannot stand in for this run, because it is contaminated. In its rounds 1–3 the four non-frontier agents had read Sonnet's and Qwen's answers. If we re-count their votes inside that debate, the apparent gain is +11 points. Of its 26 "fixes", 21 are GLM's cut-off rows reappearing and 16 are round-0 ties.

**Limits.** There are 146 contested problems, only 6 of them olympiad. The control (0 of 40 broken) cannot rule out breaking at a rate of several points on the undebated problems. The strongest-first rule and the choice of GLM as best member are in-sample. The label audit is one-sided against this pool. Mistral was rate-limited upstream, so within each round its rows were written later than its peers'.

## 6.5 Public debate traces

A second, independent data set tests whether the collapse is general. It holds 9,600 public debate traces in 16 files, each with three copies of one model debating in three all-to-all rounds, across sixteen model and task configurations. Six primary MMLU-Pro files (Gemini 3-Flash, GPT-5.4-mini, Llama-3.1-8B, Phi-4-mini, Qwen3-4B, Qwen3-8B) are joined by ten others (Table 6.3). Hypotheses and decision rules were pre-registered

before any outcome was computed; intervals are a 2,000-resample cluster bootstrap over questions. ICC is the within-problem intraclass correlation of correctness among the three agents.

**Table 6.3. Public three-agent debate traces, round 0 → round 3. The CAPA change is the q = 0 variant of accuracy-adjusted agreement, which measures agreement beyond what the agents' accuracies imply.**

| file group | files | ICC round 0 → 3 | ICC rises every round | all-wrong at round 0 corrected by round 3 | CAPA change |
|---|---|---|---|---|---|
| Primary (MMLU-Pro, one model) | 6 | 0.48–0.91 → 0.62–0.99 | 6 of 6 | Qwen3-4B 31% of 793 [28, 34]; Qwen3-8B 45% of 110 [36, 54]; other four 1–5% | +0.13 … +0.35 |
| Heterogeneous panels | 3 | 0.37–0.47 → 0.75–0.78 | 2 of 3 | 0–4% of 17–24 | +0.36 … +0.38 |
| GSM8K | 3 | 0.45–0.65 → 0.78–0.96 | 1 of 3 | 0% of 5–9 | +0.36 … +0.43 |
| Qwen3 thinking on/off | 4 | 0.59–0.72 → 0.87–0.97 | 4 of 4 | 0–10% of 45–51 | +0.20 … +0.42 |
| Self-only controls | 2 | 0.65–0.73 → 0.66–0.74 | flat | 0–4% of 46–49 | +0.00, +0.04 |

On the six primary files the ICC rises from 0.48–0.91 to 0.62–0.99, and it rises in every round, as the pre-registration required. Over all 16 files the rise is monotone on 13. The three exceptions (GSM8K Llama-4-Scout, GSM8K Mistral-Small-4 and one heterogeneous panel) dip slightly at the end. By round 3 the primary panels have an $n_{\text{eff}}$ of 1.01–1.34 out of 3. On the self-only controls the ICC stays flat (0.650 → 0.662 and 0.726 → 0.740). Because a rising ICC could reflect rising accuracy, a pre-registered follow-up computed accuracy-adjusted agreement (the q = 0 variant of CAPA, not the exact published metric). It rises by +0.13 to +0.43 on all 16 peer files and by +0.00 and +0.04 on the two self-only controls. At round 3, when two agents are both wrong they give the same wrong answer in 61–100% of cases on the peer files. An earlier analysis of these traces claimed that debate corrects "31–45% of the problems where all three agents started wrong". That holds on only 2 of the 16 files, and the 45% comes from an atypical Qwen3-8B file. The traces do reject the modal-hit rate as a cap on debate. On weak models the final majority exceeds the round-0 modal-hit rate by 6 to 20 points, with intervals clear of the pre-set 3-point band, because peers shift each problem's answer distribution.

## 6.6 Bounds

We tested two bounds, with predictions written down before the computation, on data already collected. The first is the Moore–Shannon [31] restoring-organ curve: the probability that the round-3 majority is right, as a function of how many agents were right at round 0. On the mixed panel this curve is monotone with no crossover. Problems where one or two of the six agents start right end with a correct majority 56–86% of the time, and problems with four or more right never lose. But 79% of problems start with four or more correct agents, so the corpus sits at the saturated end and the curve's threshold cannot be located.

The second is the MAP ceiling. In every arm, the six DeepSeek copies end within +0.013 of the share of problems on which DeepSeek's own 16-draw mode is correct (0.892). Same-model debate decodes the mode and no more. The mixed panel, on the same problems, ends +0.061 [+0.028, +0.098] above that ceiling, all of it on GPQA Diamond and olympiad. Those are the sets where the capped models were cut off most (Section 5.2). This is a gain over one model's ceiling, which is estimated in-sample from the same 16 draws. It is not a gain over the best member.

## 6.7 What this shows and what it does not

Debate collects one to two points where the members' errors are not shared, and nothing on a saturated pool. In every setting it loses coverage, and the panel collapses toward one voter. Stand by stand:

- **Six mixed models, 225 problems.** The majority rises from 0.924 to 0.973 under the fixed tie order (p = 0.001) and from 0.951 to 0.973 with the strongest agent breaking ties (p = 0.06). Ten of the 11 fixes are on problems where an agent was cut off at round 0 (in six of them the cut-off agent simply casts no vote), so this gain depends on the 8,192-token cap. On the 162 complete problems debate makes one fix. Against the best member it ranges from −1.8 to +2.0 points. Mean φ rises from 0.34 to 0.85, and unanimous-wrong problems rise from 0 to 4.
- **Six copies of one model.** +3.5 points, about half of it (+1.7) with no peers; the copies stay at their MAP ceiling; unanimous-wrong 6 → 22.
- **Four non-frontier models, pre-registered.** Debate collects about 1.3 of 3.6 points over the in-sample best member (p = 0.052). Coverage on the contested problems falls from 1.000 to 0.925, and $n_{\text{eff}}$ falls from 5.7 to 1.5.
- **Synthesis.** At one call per problem it lands near debate (0.849 against 0.870) and recovers 1 of 10 all-wrong problems on the non-frontier pool, against 0 of 16 on the saturated pool.
- **Public traces.** Agreement beyond accuracy rises on all 16 peer files and $n_{\text{eff}}$ falls to 1.01–1.34; correction of all-wrong problems holds on 2 of 16; the modal-hit rate is not a cap.
- **Bounds.** The restoring-organ curve is monotone, but its threshold cannot be located on a saturated corpus. Same-model debate stops at the MAP ceiling, while the mixed panel clears it by +0.061 [+0.028, +0.098].

What it does not show: the saturated-pool gains and the re-elicitation share carry no intervals; the tie rules, the strongest agent and the stopping rule were chosen in-sample, after the data were seen; there is no correction for multiple comparisons across tie rules, benchmarks and arms, and the only saturated-pool p < 0.05 uses the tie rule least favourable to the round-0 vote. A final version should fix these rules on held-out problems and report the 162 complete problems as primary. The non-frontier result rests on 146 contested problems and an in-sample tie rule. Section 13 collects these limits.

# 7. Best-of-n with a posterior check: judges, certification and the oracle

**What this section shows.** A judge panel picking among candidate answers collects a few points over the solvers' plurality vote, with the sign set by the tie rule: +8.1, +4.0 or −1.4 points on the contested items of the full pool , +3.4, +6.8 or +5.5 points in a pool without the frontier models . In neither pool does the panel clearly exceed its best judge or the best member. Certification is safe only under unanimity, which passes 1 of 54 wrong candidates, and only because one strict judge sits on the committee: unanimity of three weaker judges passes 8 of 54 . The large prize, an oracle 10 to 32 points above the best member on agent tasks, is unmeasured: we ran no partial verifier (generated tests, execution, a reranker).

## 7.1 Three operators that act after the answers exist, and their algebra

These operators act after n candidate answers exist (terms as in Section 2). In **selection** a judge (a member, possibly one of the solvers) is shown the candidates and picks one. In **certification** a judge is shown one candidate and says only whether it is correct; a committee accepts it if at least $q$ judges do. The **oracle** is a perfect selector; its accuracy is the coverage.

The oracle is a ceiling: acc(selection) ≤ oracle always holds. The assumed floor, acc(best member) ≤ acc(selection), is *not* guaranteed, because a judge can pick a wrong candidate on an item the best member gets

right. Split the N problems into three groups: $A$, items where the best member is right; $B$, items where the best member is wrong and some other member is right; $C$, items where every member is wrong. Let $r_B$ be the judge's rate of correct picks on $B$ and $w_A$ its rate of wrong picks on $A$; on $C$ no pick can be right. Then

$$\text{acc(selection)} - \text{acc(best)} = (r_B \cdot |B| - w_A \cdot |A|) / N,$$

so selection beats the best member if and only if $r_B \cdot |B| > w_A \cdot |A|$: the correct picks gained where the best member fails must outnumber the wrong picks made where it succeeds. Comparing rates is not enough: $|A|$ usually dwarfs $|B|$. On the SWE-bench Verified leaderboard [1, 1] matrix the best of the top seven systems resolves 0.792 of 500 issues and the oracle 0.890 . That makes $|B|/|A| = 0.098/0.792 \approx 0.12$, so even a judge perfect on $B$ beats the best system only if it errs on fewer than 12% of the issues the best system already solves.

## 7.2 What the literature says

A theorem on certifying committees ([32]) shows that a rule seeing only counts, identities and visible judgments cannot guarantee validity when errors are unrestricted, and that clones of one model family cannot push the unsafe-certificate probability below the family's common-mode error. Multi-Agent Verification ([33]) beats self-consistency in most settings but loses at low compute. A study of pivotal votes ([34]) finds that a verification signal replacing one judge gains only on queries one vote from flipping, with pooled $n_{eff}$ unchanged; our $n_{eff}$ carries that caveat.

## 7.3 Data and protocol: two runs

**Full pool.** Data: the text corpus of 733 problems and its six-model debate set (Section 4). An item is **contested** when the answering debate agents disagree, at least one right and one wrong. There are 75 contested items (31 olympiad, 24 GPQA Diamond, 12 GSM-Hard, 8 MMLU-Pro); all five judges saw 74 . The judges were Claude Haiku 4.5, Claude Sonnet 5.5, DeepSeek V3.2, Qwen 3.7 Plus and GLM 5.3 Flash. Each saw the distinct candidates shuffled and anonymised, each with a 700-character excerpt of a proposer's reasoning, and picked one or answered "none" (its own answer may be among them). The panel takes the plurality of picks; truncated or unparseable picks are unnamed. Three tie rules, none fixed in advance, apply to both sides: first in a fixed order, ties split evenly, strongest first (in-sample). For certification the same judges saw one candidate at a time and answered CORRECT or INCORRECT; each item also carried a **decoy**, a wrong draw no proposer had put forward. Only the judge-weighted vote prediction (Section 7.4) was pre-registered.

**Pool without the frontier models.** This is the same run as the debate of Section 6.4, among the four non-frontier models of Section 5.4 (Haiku, DeepSeek V3.2, Mistral Large, GLM-5.3-flash), on the 146 of the 733 problems on which the four disagree with at least one right and one wrong. Three of them (Haiku, DeepSeek, GLM; Mistral never judged) judged the four members' round-0 candidates with the same prompt (two candidates on 110 problems, three on 34, four on 2) . A band of ±3 points for the panel's lead over the plurality was pre-registered. As a check, the same three judges were also scored from their stored picks over the full pool's candidates; those lists contained the frontier models' candidates, so this is a contamination check, not a non-frontier operator.

## 7.4 Selection on the full pool

**Table 7.1. Judge panel on the contested items of the text corpus. Picks and solver plurality are on the same items; "first in fixed order" breaks solver ties toward DeepSeek (weakest solver here) and panel ties toward Haiku.**

| quantity | value |
|---|---|
| Pick accuracy per judge, unnamed pick = wrong (74–75 items) | Sonnet 0.80, Haiku 0.73, Qwen 0.65, DeepSeek 0.52, GLM 0.48 |

| quantity | value |
|---|---|
| Pick accuracy on named picks | Qwen 0.89 (55), Sonnet 0.87 (68), GLM 0.82 (44), Haiku 0.75 (73), DeepSeek 0.67 (58) |
| Mean pairwise ϕ of "pick correct", 74 items; $n_{eff}$ of 5 | 0.16 (tetrachoric 0.27), 3.07 with unnamed = wrong; 0.29, 2.31 pairwise-complete |
| Panel vs solvers' plurality, first in fixed order | 0.838 vs 0.757 (+8.1 pts; 11 won / 5 lost, McNemar p = 0.21) |
| Panel vs solvers' plurality, ties split evenly | 0.842 vs 0.802 (+4.0 pts) |
| Panel vs solvers' plurality, strongest first (Sonnet both sides) | 0.851 vs 0.865 (−1.4 pts; 4 / 5, p = 1.00) |
| Strict majority of judges (≥ 3 of 5 on one candidate) | answers 54 of 74; right on 0.944 [0.85, 0.98]; solvers' plurality on those 54: 0.796 |
| Whole set, 235 items: plurality / plurality + panel | 0.915 / 0.940 (first-judge ties) or 0.953 (tied panel falls back to plurality) |
| Best single solver, truncation = wrong (in-sample) | Sonnet 0.945 |

Like for like, Haiku picks better than it solves (0.75 against 0.56) and so does DeepSeek (0.67 against 0.45); Sonnet, Qwen and GLM pick about as well as they solve, within 2 points . The panel's lead over the solvers' plurality runs from +8 to −1 points with nothing changed but the tie rule. Under the fixed order, 9 of the panel's 11 wins are items where the solvers tied and the order gave the tie to a wrong answer; on the 63 contested items with no solver tie the plurality is 0.873 and the panel 0.825 . A strict majority of judges is usually right where it exists (0.944 on 54 items) but abstains on the other 20. Over all 235 items the panel moves the vote by 2.5 to 4 points, to within a point of the best solver (0.945).

**Judge-weighted vote.** In a related operator each judge model's own answers are weighted by its pick accuracy. Over 1,017 text-corpus items this lifts the five-model vote from 0.864 to 0.895, +0.015 [+0.000, +0.029] over the best of those five; with all seven models (the two non-judges at the mean weight) it is −0.015 [−0.028, −0.002] below the best single . Of the pre-registered prediction, “within ±0.01 of plurality” was falsified and “no win over the best single with an interval excluding zero” held. The weights come from an earlier judge pass (81 contested items for Sonnet) on the same problem sets, so +0.015 is an upper bound.

*Caveats*. 74–75 contested items, 11 versus 5 discordant, one corpus; best solver, strongest tie-breaker and weights are in-sample; tie rules chosen after the data.

## 7.5 Selection in the pool without the frontier models

Pick accuracy, an unnamed pick scored wrong: DeepSeek 0.829, Haiku 0.644, GLM 0.452 (0.868 on the 76 it named). The pick errors correlate at $\bar{\varphi}$ 0.20, $n_{eff}$ 2.1 of 3. DeepSeek picks 0.829 while solving 0.575 of the same problems, the largest pick-over-solve gap here, and alone answered "none" on an all-wrong problem (2 of 10) .

The panel's lead over the four members' round-0 plurality is +3.4 (first in fixed order), +6.8 (ties split) and +5.5 points (strongest first: 0.856 against 0.801; 17 won / 9 lost, $p$ = 0.17), so the pre-registered ±3 band was missed. Its strict majority answers 116 of 146 at 0.871 (plurality 0.845). On the 733 problems the result lands between 0.939 and 0.958 depending on the tie rule, against GLM's in-sample 0.950, so the sign against the best member is again set by the rule. Against the three-round debate on the same 146 (Section 6.4) it is 0.856 against 0.870 (8 / 10, $p$ = 0.82); one DeepSeek synthesis call reaches 0.849.

**Contamination check.** The same three judges over the full pool's candidates do not beat the six solvers' plurality under any rule (0.720 against 0.747, 0.792 and 0.853) . The lead comes from weak solvers leaving

room, not from judges good in general: a few points of headroom, about what debate collects at a third of the calls. *Caveats:* 146 problems, 26 discordant; strongest-first orders and the pool itself were chosen in-sample.

## 7.6 Certification and peer disagreement

**Table 7.2. Certification: committee of five judges over the 94 candidates all five ruled on (40 correct, 54 wrong; 20 of the wrong are decoys; 48 of the 94 are GSM-Hard). A judge counts as accepting if it accepted any format of the answer.**

| accept if ≥ q of 5 | accepted | precision | recall of correct | wrong accepted (of 54) | if every format must be accepted: wrong / recall |
|---|---|---|---|---|---|
| 1 | 66 | 0.61 | 1.00 | 26 | 26 / 1.00 |
| 2 | 56 | 0.71 | 1.00 | 16 | 16 / 1.00 |
| 3 | 49 | 0.82 | 1.00 | 9 | 8 / 0.95 |
| 4 | 44 | 0.86 | 0.95 | 6 | 5 / 0.82 |
| 5 | 35 | 0.97 | 0.85 | 1 | 0 / 0.68 |

Sonnet falsely accepts 0.06 of 189 wrong candidates and no decoy in 64; the other four falsely accept 0.23–0.43 of wrong candidates and 0.09–0.24 of decoys. Every judge favours its own wrong answer: Sonnet 0.36 (n = 14), DeepSeek 0.56 (32), GLM 0.62 (8), Haiku 0.86 (35), Qwen 1.00 (5) ("own" = equal to its matrix answer, not necessarily recognised). False accepts correlate at mean φ 0.38 over the 54 wrong candidates, an $n_{eff}$ of 1.98 for five judges. Only unanimity has a point false-accept rate under 5%: it passes 1 of 54 wrong candidates (1.9%, Wilson 95% [35] [0.3%, 9.8%]) at recall 0.85, and 0 of 54 (upper bound 6.6%) at recall 0.68 if every format must be accepted; four of five passes 6 of 54 (11%) at recall 0.95. That safety rests on the two strictest judges, Sonnet above all: unanimity of Haiku, DeepSeek and GLM alone passes 8 of 54 (0.148 [0.077, 0.266]) . *Caveats:* 54 wrong candidates, half GSM-Hard and 37% planted decoys.

**Peer disagreement as an error detector.** The target is DeepSeek's matrix answer on 225 debate items, wrong on 0.169 of them. The number of the five other models that disagree with it ranks its errors at AUROC 0.981; disagreement among its own temperature-1 draws reaches 0.892, 0.902 and 0.932 with 3, 5 and 15 draws . Five peers beat fifteen own draws by +0.049 [+0.009, +0.101], but against one strong peer (Sonnet alone, AUROC 0.931) the gap is −0.001 [−0.065, +0.068]: the peers are mostly better solvers. One target model only.

## 7.7 The oracle on agents and what an imperfect verifier collects

On the SWE-bench Verified matrix [1] the majority of the top seven systems resolves 0.794 of 500 issues, +0.002 [−0.022, +0.010] over the best system's 0.792, while the oracle reaches 0.890 : a prize of 9.8 points that voting does not collect. Below the top, in the accuracy bands 0.4–0.7, the prize is 19 to 32 points (Section 5.5) , and on τ-bench pass@4 exceeds pass@1 by 14 to 25 points (Section 11.3) . The hidden test suite is a perfect selector; between it and voting lie partial verifiers (reproduction tests, execution, LLM rerankers). The 82 submissions that declare an internal best-of-n selector match their peers in resolved rate, φ and union-minus-best gap (Section 5.5) , but their rates are post-selection, so they do not measure what a selector recovers of its own candidates' union. Two released candidate sets do, at no model calls of our own.

*A model verifier on SWE-bench Verified.* The R2E-Gym release [36] gives, for its 32B agent, the hidden-test outcome of 12,695 candidates on the 500 bugs (19 to 26 per bug) and, for each candidate, the probability of "yes" from the authors' 14B execution-free verifier . pass@1 is 0.319 [0.286, 0.351], pass@*n* 0.644 [0.598, 0.684], a prize of 32.5 points on 299 contested bugs. The verifier has AUROC 0.805 and no operating point

with both errors small: at threshold 0.5 $\alpha = 0.31$, $\beta = 0.23$; at 0.7 $\alpha = 0.05$, $\beta = 0.76$. Table 7.5 gives what it collects.

**Table 7.5. Selection among *n* random candidates per bug on the R2E-Gym release [36] (100 subsets per bug; 95% bootstrap intervals over 500 bugs, 1,000 resamples). Share = (accuracy – random candidate) / (oracle – random candidate) at the same *n*.**

| Rule | *n*=2 | 4 | 8 | 16 | 25 |
|---|---|---|---|---|---|
| oracle, pass@*n* | 0.409 | 0.483 | 0.552 | 0.611 | 0.642 |
| random candidate | 0.320 | 0.319 | 0.319 | 0.319 | 0.320 |
| model verifier, argmax: accuracy | 0.365 | 0.391 | 0.408 | 0.423 | 0.428 |
| share of the prize | 0.51 [0.46, 0.56] | 0.44 [0.39, 0.49] | 0.38 [0.32, 0.44] | 0.36 [0.29, 0.42] | 0.33 [0.26, 0.41] |
| model verifier top 3, then generated-test score (the authors' rule): share | 0.33 [0.27, 0.39] | 0.41 [0.36, 0.46] | 0.42 [0.38, 0.47] | 0.40 [0.34, 0.45] | 0.39 [0.33, 0.45] |
| threshold 0.5: share | 0.35 [0.31, 0.39] | 0.31 [0.26, 0.34] | 0.26 [0.22, 0.30] | 0.22 [0.18, 0.26] | 0.20 [0.16, 0.24] |
| threshold 0.7: share | 0.11 [0.07, 0.14] | 0.13 [0.10, 0.17] | 0.15 [0.11, 0.18] | 0.17 [0.13, 0.22] | 0.18 [0.13, 0.24] |

Three things follow. First, accuracy rises with $n$ while the share falls, from half at two candidates to a third at sixteen for the argmax rule (share(16) − share(4) = −0.08 [−0.12, −0.04]): the distance to the oracle widens, as the precision limit of Section 2.2 says. At strict thresholds the share rises instead (+0.04 [+0.01, +0.07] at 0.7), because with $\beta$ of 0.76 more candidates still heal false rejects faster than false accepts hurt; the best threshold rises with $n$, from 0.35 at $n$=2 to 0.6 at $n$=16. Second, the verifier's errors are clustered by bug: the intraclass correlation of accept verdicts within a bug is 0.26 to 0.43. The formula of Section 2.2 with one pooled ($\alpha$, $\beta$) overpredicts the threshold rule by 3.8 points at threshold 0.6 and 5.7 at 0.7 ($n$=12, held-out half); per-bug rates estimated on one half of each bug's candidates and applied to the other half remove 0.36 and 0.65 of that error, and verdicts regenerated as independent coins from the per-bug rates give the same removal within 0.08, so what split-half does not remove is estimation noise from about 13 candidates per half, not structure the per-bug model misses. Third, the blind spot has a side: against the item-response difficulty of Section 11.2 (fitted on the 175 leaderboard submissions, 474 bugs matched) the per-bug false-reject rate correlates at +0.46 [+0.36, +0.55] (Spearman; 0.47, 0.72, 0.75 by difficulty tercile at threshold 0.6) and the false-accept rate at +0.09 [−0.01, +0.18]. This verifier rejects correct patches on hard bugs; it does not accept wrong ones more often there.

*An agent's own tests.* The SWE-rebench OpenHands rollouts [37] (67,074 rollouts of Qwen3-Coder-480B on 3,792 issues, the benchmark's resolved flag on every row, and on 47% of rows whether the agent's own generated tests pass its own patch) give the execution-based case. pass@1 is 0.476, pass@$n$ 0.601, a prize of 12.6 points. "Own tests pass" has $\alpha = 0.69$, $\beta = 0.24$ and AUROC 0.53: the agent's tests pass its patch on 72% of rows whether or not the patch is right, only 30% of its test sets are valid against the reference fix, and the rule collects 0.02 [−0.01, 0.05] of the prize at $n$=2 and 0.04 [0.01, 0.06] at $n$=8. "Submitted a patch" has $\alpha = 0.85$ and loses share with $n$ (−0.015 [−0.026, −0.004] from 4 to 8 on the same 4,737 issues). The one rule with small $\alpha$, "own tests pass and are valid against the reference fix" ($\alpha$ 0.12, $\beta$ 0.63), gains share with $n$ (+0.032 [+0.007, +0.057]), but it uses the hidden fix and is not a deployable verifier. The verifier that collects the prize on agent tasks must have a small false-accept rate, and tests written by the solver for its own patch do not.

*Our own verifiers on the same candidates.* To put a model judge and model-written tests on identical candidates we took 200 contested SWE-rebench issues with a public environment image, 8 Qwen3-Coder rollouts each (1,600 candidates; pass@1 0.536, pass@8 0.960, a prize of 42.4 points on these issues), and ran three verifiers in cloud containers, pre-registered : Claude Haiku as a judge reading issue and patch; tests

written by Claude Haiku from the issue text alone; and, in a second pre-registered run, tests written by Claude Haiku with read-only access to the repository at the base commit. Table 7.6 gives the error rates and what each collects. The judge at threshold 0.5 has $\alpha$ 0.20 [0.15, 0.26] and $\beta$ 0.54 [0.48, 0.60], AUROC 0.68, and collects 0.15 of the prize at eight candidates by threshold and 0.19 by argmax (0.32 at two); its errors are per-issue (intraclass correlation 0.65) and on the reject side ($\beta_i$ against difficulty +0.39 [+0.26, +0.51]), as for the R2E verifier. Tests written without reading the code are valid against the reference fix on 98 of 200 issues and have $\alpha$ 0.39 [0.31, 0.47], $\beta$ 0.48 [0.40, 0.55]; tests written after reading the code are valid on 163 issues, $\beta$ falls to 0.25 [0.19, 0.31] and the AUROC rises from 0.57 to 0.65, but $\alpha$ does not fall, 0.46 [0.39, 0.53]: a patch the hidden tests reject usually does fix the behaviour the issue names and fails elsewhere. The reading tests collect 0.34 [0.27, 0.42] of the prize at two candidates and 0.23 [0.16, 0.30] at eight; used as a gate before the judge's ranking they collect 0.42 [0.31, 0.52] and 0.28 [0.15, 0.40], the largest share of any rule in this paper, and within 0.04 of the same tests restricted to those valid against the reference fix. The pre-registered expectation that execution-based tests would be strict (small $\alpha$) was wrong for model-written tests on this benchmark, in both runs; what they supply is recall.

**Table 7.6. Verifiers on the same 1,600 Qwen3-Coder candidates, 200 contested SWE-rebench issues (95% cluster-bootstrap intervals over issues). Share = share of the prize on these issues at the threshold rule unless marked argmax.**

| Verifier | $\alpha$ | $\beta$ | AUROC | Share, $n$=2 | $n$=8 |
|---|---|---|---|---|---|
| Judge (Claude Haiku), threshold 0.5 | 0.20 [0.15, 0.26] | 0.54 [0.48, 0.60] | 0.68 | 0.17 | 0.15 [0.08, 0.22] |
| Judge, argmax | | | | 0.32 [0.22, 0.41] | 0.19 [0.07, 0.31] |
| Tests written from the issue text, strict | 0.39 [0.31, 0.47] | 0.48 [0.40, 0.55] | 0.57 | 0.15 | 0.11 [0.06, 0.17] |
| Tests written after reading the repository, strict | 0.46 [0.39, 0.53] | 0.25 [0.19, 0.31] | 0.65 | 0.34 [0.27, 0.42] | 0.23 [0.16, 0.30] |
| Reading tests as gate, then judge argmax | | | | 0.42 [0.31, 0.52] | 0.28 [0.15, 0.40] |
| Tests by Qwen3-Coder-30B from the issue text, strict | 0.11 [0.06, 0.16] | 0.86 [0.80, 0.91] | 0.50 | 0.06 | 0.04 [0.01, 0.08] |

*Caveats*. The R2E look was a first analysis; the intervals, the split-half test and the difficulty correlations were then pre-registered and run on the same data, so the hypotheses are not independent of it. One agent and one verifier per data set; the R2E release carries no licence statement; hidden tests are themselves imperfect. The 200 issues of Table 7.6 were chosen as contested on the hidden labels, so the shares there are shares on contested issues (the whole-benchmark share is 0.29 of them) and the error rates are candidate-level quantities unaffected by that choice; the test writer used one model and at most five tests per issue; the Claude calls ran under a subscription. What is still unmeasured: a verifier whose false-accept rate is small without the hidden tests, which none of the model-written tests here supplied. A Good–Toulmin [38] forecast of pass@16 from 4 draws lands within 0.03 of the observed value but is biased upward by about that much, and its interval under-covers . *Caveats:* the top seven and the bands are cut on the issues they are scored on, and submissions are not independent agents (Section 11).

## 7.8 What this shows and what it does not

It shows that a judge panel collects a few points over the solvers' plurality, with the sign set by the tie rule (+8.1 to −1.4 points on the full pool, +3.4 to +6.8 without the frontier models), lands within a tie rule of the best member (0.940–0.953 against 0.945; 0.939–0.958 against 0.950), and never clearly exceeds its best judge. As

pickers the judges are fairly independent ($n_{eff}$ 2.3–3.1 of 5), as certifiers their false accepts are correlated ($n_{eff}$ 2.0) and self-preference dominates; task, item mix (16% versus 51% GSM-Hard) and truncation scoring are confounded there. Certification is safe only by unanimity (1 of 54) and only because of one strict judge (the three-judge committee without Sonnet and Qwen passes 8 of 54; and even the five-judge rate's interval reaches 9.8 %). The weighted vote adds at most +0.015 over the best of five; peer disagreement detects errors no better than one strong peer. It does not show that any panel beats the best member. On the one large prize, the 10 to 32 points of oracle on agent tasks, it shows on released and on our own candidates that a model verifier collects half of it at two candidates and a third at sixteen, that its errors are per-task and sit on the reject side, that tests written by a model accept 39 to 69% of wrong patches whether written by the solver, from the issue text or after reading the code, and that the best deployable rule, model-written tests as a gate before a ranking judge, collects 0.28 of the prize at eight candidates; a verifier with a small false-accept rate that does not use the hidden tests remains to be found. The algebra of Section 7.1 binds: a judge must be nearly flawless on the items the best member already solves.

# 8. Routing and orchestration: best-of-n with a prior check

**What this section shows.** We ran no router, so routing has no measurement of our own; this section bounds it and reads the published routers against the bound. On the text corpus (Section 4; 738 rows, one temperature-0 answer per model) the best member scores 0.973 and the oracle 0.997, so a router over our pool can add at most about 2.4 points; a route that picks one model per problem set, chosen in-sample, scores 0.979 and collects 0.7 of them, a share of about 0.27. Among published routers that report both ends, Trinity collects about 0.15 of its gap (69.1 to 70.4 against an oracle of 78.5) and Chen's deployable routers about 0.09 (interval −0.67 to 0.50); the Conductor and Fugu report no oracle, so their share cannot be computed. Both collect a minority of their gap to the oracle, and neither exceeds its oracle.

## 8.1 The operator

This section asks how much an operator that chooses a member *before* answering can add over the best member, first as a bound, then in published systems and on our corpus. A *router* is an orchestrator that, for each problem, chooses which member will answer before any answer exists. It acts on a prior: what it has learned about which member tends to be right on problems of this kind. We call this a *prior check*, to contrast it with *selection* (Section 7), where a judge chooses among finished answers and so checks after the fact. Trained orchestrators add roles, multi-turn workflows or subtask prompts; we keep "router" for the part that picks a member.

**Formal statement.** Let member $i$ answer problem $q$ correctly with probability $p_i(q)$, and let a router send $q$ to one member $r(q)$, chosen without seeing any answer. Its expected accuracy is the mean over problems of $p_{r(q)}(q)$, so its attainable ceiling is

$$\text{acc(router)} \le R^*, \quad \text{where } R^* = (1/N) \cdot \Sigma_q \, [\, \max_i p_i(q) \,]$$

(for each problem $q$, first the best member's probability of being right on that problem; then the average over the $N$ problems: the ceiling of a router that always knows whom to ask), and

$$R^* \le \text{oracle}, \quad \text{where oracle} = (1/N) \cdot \Sigma_q \, \text{P(at least one member correct on } q)$$

(the average over problems of the chance that *some* member is right, which is never below the best member's chance).

The second line is the oracle of the pool (Section 2). For deterministic members $R^*$ and the oracle coincide. For stochastic members $R^*$ is lower: the union of single runs counts lucky draws that no prior choice can target, so an oracle computed from one run per member overstates what a router can reach. The bound holds for a router that sends the problem to one member once. A multi-turn orchestrator that switches members between turns, or shows one member's output to the next, is not bounded by the single-run union; for it the oracle is an empirical reference, not a structural limit. And the gain over the best member, $\Delta$ = acc(router) − acc(best member), is bounded by the gap "oracle − best member", which is small whenever the pool's members fail on the same problems. A router only rearranges answers the members would already give in expectation, adding no computation of its own, so it is bounded like plurality voting, by the gap that the members' shared failures on hard problems leave open (Sections 5.3–5.5).

The bound therefore says where to look: at the gap between the oracle and the best member, and at the share of it a router collects. Section 8.2 reads the published orchestrators against it.

### 8.2 What the literature reports

On their own tables, the trained orchestrators beat the best single model in their pools by one to nine points.

- **Trinity** ([39]) trains a coordinator of about 20,000 parameters on a 0.6B backbone to choose, at each of up to five turns, one of seven models and one of three roles (planner, worker, verifier). All models are capped at 4,096 output tokens with minimal reasoning effort. It averages 70.44 against 69.08 for the best fixed model per benchmark under the same protocol, a margin of 1.4 points. Trinity is the only paper in the family that reports the per-question oracle of its pool, and it stays below it on every in-distribution task (MATH500 0.88 vs 0.91, MMLU 0.92 vs 0.97, RLPR 0.41 vs 0.61, LiveCodeBench 0.61 vs 0.65). Whether that oracle uses one seed or several is not stated, so part of the 20-point RLPR gap may be sampling variation rather than misrouting. Its routing is close to per-task: 87.9% of LiveCodeBench calls go to GPT-5 and 82.2% of MATH500 calls to Gemini.
- **The Conductor** ([40]) is a 7B model trained with GRPO to write workflows of up to five steps, each with a subtask prompt, a worker and an access list of earlier outputs. Its answer is built from several workers' output, so the router bound does not apply to it. In the capped setting it is 3.9 points above the best worker per benchmark in-distribution and 9.4 points above on held-out tasks (61.93 vs 52.57). The held-out figure is above what any one-model-per-task router could reach on that table. About 6.7 of the 9.4 points come from thirty AIME 2025 problems, and the workers are cut to short answers: Claude Sonnet 4 solves 35.3% of AIME under the cap against 74.3% uncapped. The paper reports no per-question oracle, so whether the composed answers exceed the pool's union is not shown.
- **Fugu** ([41]) routes among three named frontier models (Gemini 3.1 Pro, Claude Opus 4.8, GPT-5.5), choosing a worker at every turn in multi-turn tasks. **Fugu-Ultra** applies the Conductor recipe to the same pool. Fugu-Ultra and Fugu exceed their best worker on several rows (GPQA Diamond 95.5 vs 94.3; Terminal Bench 82.1 and 80.2 vs 78.2). On other rows they score below it, for example plain Fugu on SWE-Bench Pro (59.0 vs 69.2) and on MRCRv2 (86.6 vs 94.8). Most baselines are provider-reported, from other harnesses, without intervals, and on multi-turn tasks the executor changes mid-trajectory; the report does not separate routing, harness and run variance, and reports no oracle, majority vote or fixed-worker baseline. The release's parity claim with Fable 5 rests on such baselines and on a pool we describe as the previous generation's public models; Fugu-Ultra is above the Fable 5 / Mythos column on five of eight benchmarks, by 0.5 to 3.4 points, and below on three, by 1.5 to 6.3. At answer time, all three trained systems rely on the trained prior rather than on an external check.
- **Ceiling papers.** Chen ([15]) frames the all-wrong rate as the binding ceiling on routing, voting and mixture-of-agents. In that study, deployable routers collect about 9% of the oracle gain (interval −0.67 to 0.50 of the gain), and an LLM router picks the single best model on every query. [26] shows coverage rising toward 1 while majority voting plateaus, the same gap between "some draw is right" and "a chosen

draw is right". AB-MCTS ([42]) selects with a partial checker and belongs with the partial verifiers of Section 7.7.

In short, the trained orchestrators report margins of one to nine points over their best worker; the one that reports an oracle stays below it, and the others report none, so their margins cannot be read as a share of the available gap. Section 8.3 computes the same quantities on our own data.

## 8.3 What our data can and cannot test

We ran no router. What the text corpus (Section 4) allows is ceiling arithmetic. It holds one temperature-0 answer per model and problem for seven models (738 rows; 733 answered by all seven). Table 8.1 recomputes the quantities from per-problem-set accuracies rounded to three decimals and weighted by problem count; the gap between the oracle and the best member is 2.4 points from the rounded totals and 2.5 from the unrounded sums.

**Table 8.1. Router ceiling arithmetic on the text corpus (738 rows; one temperature-0 answer per model). All choices are in-sample.**

| quantity | accuracy | problems (of 738) |
|---|---|---|
| best member (Claude Opus, pooled over all four problem sets) | 0.973 | ≈ 718 |
| "one model per problem set" route (Opus on MMLU-Pro, GLM on GPQA Diamond, Sonnet on GSM-Hard and Olympiad) | 0.979 | ≈ 723 |
| oracle (at least one of seven correct) | 0.997 | ≈ 736 |
| plurality of the seven (for reference) | 0.957 | ≈ 706 |

A router over this pool can add at most about 2.4 points over the best member. A per-problem-set route collects 0.7 of them, about five problems out of 738. Both the best member and the per-set route are chosen in-sample. On the 91 GPQA and 29 Olympiad problems the per-set winner is ahead by one problem each, so a held-out choice may collect less. On the 733 problems answered by all seven, the per-set route is 0.981 and the oracle 0.999. The joint-failure rate is about two problems in 738, so the pool is near its ceiling by construction. A small headroom here says little about pools selected for complementarity, as Trinity's was.

We cannot test the parity claim, any of Fugu's numbers, multi-turn routing, learned selectors, or composition of answers from several workers; our selectors (Section 7) are prompted judges of finished answers. Because absolute headroom differs so much between our pool and the published ones, Section 8.4 compares them on a common scale.

## 8.4 A fair comparison across studies

Absolute gains are not comparable across pools. A pool at 97% has 2.4 points to give, while a capped pool at 70% has ten or more. The comparable quantity is the gap "oracle – best member" and the share of it an operator collects, the collected share $s$ of Section 2.

**Table 8.2. Gap to the oracle and share collected, where both ends are reported.**

| study | best member | oracle | operator | share of gap |
|---|---|---|---|---|
| this corpus, per-set route (in-sample) | 0.973 | 0.997 | 0.979 | ≈ 0.27 |
| Trinity, 4-task average (best fixed model per benchmark) | 69.1 | 78.5 | 70.4 | ≈ 0.15 |
| Chen 2026, deployable router | reported as share only | | | ≈ 0.09 (−0.67 to 0.50) |
| Conductor, Fugu, Fugu-Ultra | no oracle reported; share not computable | | | |

The Trinity baseline is itself a per-task route and its oracle may be a single-seed union. With that said, the studies agree: routers collect a minority of the gap to the oracle, and no reported router exceeds it.

### 8.5 What this shows, its limits, and two zero-cost tests

This section has shown, without a routing measurement of our own, that a single-run router over our pool can add at most about 2.4 points over the best member, that an in-sample per-problem-set route collects 0.7 of them (a share of about 0.27), and that the published routers that allow the computation collect a minority of their gap (Trinity about 0.15, Chen's deployable routers about 0.09) and Trinity, the only one that reports its oracle, stays under it; the Conductor and Fugu report no oracle. A single-run router spends one member's answer per problem plus its own choice, so the equal-budget comparisons of Section 10 do not constrain it; its limit is the gap to the oracle. The larger gap on agent tasks is open to a posterior check with an external verifier (Section 7.7), not to a prior check. Section 8.3 is arithmetic on a matrix collected for other purposes, not pre-registered. Two tests need no model calls. First, an offline per-problem-set router chosen by cross-validation: choose the member on half of each problem set and score it on the other half, repeated over splits. A simple learned router on problem text fits the same protocol. This replaces the in-sample 0.979 with a held-out estimate. Second, a stochastic ceiling: two members, DeepSeek V3.2 and Claude Haiku, have up to 16 temperature-1 samples on about 60 problems per set. From these, $\text{mean}_q \max_i \hat{p}_i(q)$ can be estimated and set against the union of single draws, which measures how much a single-run oracle overstates a router's attainable ceiling. The limits these tests would not remove are collected in Section 13.

## 9. Decomposition: splitting the problem into parts

**What this section shows.** In a pre-registered run on 97 problems of the text corpus ($5.73 in total), decomposition at equal token budget added nothing on problems the model cannot solve whole: on DeepSeek's 14 numeric hard-true problems split and single pass@4 are both 0.21, Δ 0.00 [−0.29, +0.29], and none of the 7 never-solved problems was solved. It gained where one whole pass is cut off by the token cap: +0.34 [+0.18, +0.50] on GLM's 38 hard-cap problems, consistent with a budget effect, though the mechanism is not isolated. Where the model was already fine it cost accuracy (DeepSeek, medium and easy problems, −0.41 [−0.60, −0.21]). Handing the subtasks to other models (team) was never better than splitting within one model beyond noise, and at equal money the single model ties or wins on every DeepSeek stratum but one, while GLM's split attempt, which uses fewer tokens, is better at both spends. On the SWE-bench Verified leaderboard [1, 1], role teams against free agents on the same base model are mixed.

### 9.1 The operator

**Decomposition** answers a problem in three steps: a member writes a plan that lists subtasks without solving them; each subtask is solved in its own call, which sees the problem, the subtask list and the earlier results; and a final call assembles the parts into one answer. We study two versions. In **split**, one model writes the plan, solves every subtask and assembles. In **team**, the base model plans and assembles, but subtask $i$ goes to worker $i$ mod 3 from three other models (Qwen 3.7 plus, Mistral Large, Kimi K2.5). The comparison that matters is at **equal budget**: a whole decomposition attempt, with all its calls together, may use no more than the 8,192 completion tokens allowed to one whole attempt by the same model (the token cap of the text corpus, Section 4).

Formally, let $s_1, \ldots, s_m$ be the subtasks returned by the plan call $P(x)$, let $y_j = W_j(x, s_{1..m}, y_{1..j-1})$ be the worker outputs, and let the answer be $A(x, y_{1..m})$, subject to $\text{tokens}(P) + \Sigma_j \text{tokens}(W_j) + \text{tokens}(A) \le 8{,}192$. The gain is reported as $\Delta = \text{pass@}k(\text{decomposition}) - \text{pass@}k(\text{single model})$, where pass@$k$ is the share of problems solved by at least one of $k$ attempts. Here the "best member" is the base model itself, so no in-sample selection

of a member is involved. Section 9.2 reviews how often the literature makes this equal-budget comparison; Sections 9.3 and 9.4 make it on the text corpus.

## 9.2 What the literature claims

Decomposition is common in the literature, but a comparison at matched compute is rare. The Conductor ([40]) trains a model to write subtask prompts for worker models and reports nine points over its best worker on held-out tasks, with about three worker calls per question and no single model given the same compute. Fugu-Ultra ([41]) extends it to five-step workflows. CodeR ([43]) gains from roles on SWE-bench Lite but reports no single agent with the same tools at the same cost. CooperBench ([44]) reports only aggregate rates: two agents given separate features of one task reach 26–28% against 47–49% for one agent doing both, with budgets unmatched. The Interaction Tax ([45]) runs ten configurations at one budget and finds the planner–executors–integrator design the worst of them.

The SWE-bench Verified leaderboard [1, 1] data (Section 4) give a zero-cost check. We matched role-team systems to free single agents on the same base model. MASAI (GPT-4o, 0.326) is above every free GPT-4o agent (0.232–0.270). Aime-coder (Claude 3.7 Sonnet, 0.664) is 3 points above its only same-model peer (0.632). Composio SWEkit (Claude 3.5 Sonnet, 0.406) sits inside the band of its 12 peers (0.336–0.634). The best non-team GPT-4o entry, Agentless at 0.388, is itself a fixed decomposition pipeline, so it cannot serve as the "single" baseline. All three role systems declare two or more attempts, and submission dates differ. On accuracy the picture is therefore mixed and confounded. On failures it is clearer: $\varphi$ between a decomposed system and a free agent on the same model (0.42–0.67) matches the band between any two scaffolds.

Neither the published studies nor the leaderboard compare decomposition with the same model at equal budget on the same problems. The protocol below is built to make that comparison.

## 9.3 Protocol

We drew 97 problems (AIME-style olympiad, GPQA Diamond, GSM-Hard) from the text corpus of 733 problems. The sample was stratified by how often a base model had solved each problem in its 16 whole attempts in the corpus. Truncated attempts count as wrong. A problem is *hard* if it was solved in at most 4 of 16 attempts. At selection, before any call, the hard problems were split by the cause of failure. We call a hard problem *hard-true* when its failures are not explained by truncation, and *hard-cap* when at least half of its attempts were cut off by the cap. For DeepSeek v3.2 the stratum is hard-true: 20 problems (10 olympiad, 6 GPQA, 4 GSM-Hard), so 14 numeric and 6 multiple-choice, with at most one truncated attempt on any problem. For GLM-5.3-flash the stratum is hard-cap: 38 of its 41 hard problems had at least 8 of 16 attempts cut off by the cap. The remaining 3 are hard-true. Each model also has 12 medium and 6 easy problems, sampled at random. Split and team ran once on all 97 problems, and three more times on the 61 hard ones. That makes 560 runs, at temperature 1, on 9–10 October 2026.

Hypotheses, predictions and decision rules were committed before the first call. The protocol was revised twice before the main run, and both revisions are recorded with their reasons. First, after an aborted start in which GLM spent each call's cap on hidden reasoning and returned no text, reasoning was bounded to half of each call. Second, after a first wave of 186 runs in which both models solved the problem inside the "plan, do not solve" call, the planning prompt was rewritten and reasoning on that call was cut to 64 tokens. The 186-run wave is kept as a pilot and not pooled. The main runs cost $5.07 and the pilot $0.66, $5.73 in total. Single-model baselines are the 16 corpus attempts. Intervals are 95% bootstrap over problems.

## 9.4 Results

We report the gain stratum by stratum (Table 9.1), then team against split, then the same comparison at equal money (Table 9.2). In the tables p@*k* abbreviates pass@*k*.

**Table 9.1. Pass rates by base model and stratum (n = problems). The single model's p@1 is the mean over 16 corpus attempts. Its p@4 is "solved by at least one of the first 4 attempts". Split and team p@4 use the 4 decomposition attempts on hard problems.**

| Base | Stratum | n | Single p@1 / p@4 / p@16 | Split p@1 / p@4 | Team p@1 / p@4 |
|---|---|---|---|---|---|
| DeepSeek | hard-true | 20 | 0.05 / 0.15 / 0.40 | 0.20 / 0.35 | 0.15 / 0.20 |
| DeepSeek | medium | 12 | 0.62 / 1.00 / 1.00 | 0.25 / — | 0.25 / — |
| DeepSeek | easy | 6 | 0.99 / 1.00 / 1.00 | 0.50 / — | 0.67 / — |
| GLM | hard-cap | 38 | 0.10 / 0.37 / 0.58 | 0.45 / 0.71 | 0.42 / 0.66 |
| GLM | hard-true | 3 | 0.04 / 0.00 / 0.67 | 0.33 / 0.33 | 0.00 / 0.67 |
| GLM | medium | 12 | 0.67 / 1.00 / 1.00 | 0.75 / — | 0.75 / — |
| GLM | easy | 6 | 1.00 / 1.00 / 1.00 | 1.00 / — | 1.00 / — |

**Where the model cannot solve the whole problem.** On DeepSeek's 14 numeric hard-true problems, split pass@4 is 0.21 and single pass@4 is 0.21: Δ 0.00 [−0.29, +0.29]. Eight split attempts per problem add no problem to the single model's pass@16 (union gain 0.00 [0, 0]). Of the 7 numeric problems the model never solved in 16 attempts, 0 were solved in 8 split attempts each. On the 6 GPQA problems, split "solves" 3 of 5 never-solved items. Its per-attempt rate on them is 0.23 over 40 attempts, which is the 25% guessing rate for four options.

**Where the single attempt is cut off by the cap.** On GLM's 38 hard-cap problems, split pass@4 is 0.71 against single pass@4 0.37: Δ +0.34 [+0.18, +0.50]. That is above the model's own pass@16 (0.58). With the unbiased pass@4 estimator the single baseline is 0.30, so the gain is larger, not smaller. The pre-registered "signal of something more" for this stratum was split pass@1 above single pass@4. It was met: 0.45 against 0.37. The result is driven by GPQA, where the per-attempt rate on never-solved items is 0.47 (30 of 64), well above chance. On numeric items the union gain over single pass@16 is +0.09 [0, +0.23].

**Where the model was fine.** On medium and easy problems GLM gains +0.06 [−0.13, +0.22]. DeepSeek loses 0.41 [−0.60, −0.21] (pass@1 0.74 → 0.33). It solves inside the plan call and overruns its subtask calls: 62% of its split subtask calls are truncated, against 22% for GLM. In 34 of 98 runs it re-solves from scratch in the assembly, which then truncates without a final answer and is scored wrong.

**Team against split.** Handing the subtasks to other models is never better than splitting within one model, except on GPQA, where the difference is within noise. DeepSeek hard: −0.15 [−0.30, 0.00]. GLM hard-cap numeric: −0.18 [−0.41, 0.00]. GLM GPQA: +0.12 [−0.13, +0.38].

**Table 9.2. Equal money. The cost of one split attempt in whole single draws comes from the OpenRouter billing records. The single model's pass@*k* at the same spend is the unbiased estimator over its 16 draws.**

| Base / stratum | Split ÷ draw (tokens) | Split, 1 attempt | Single at same $ | Split, 4 attempts | Single at same $ |
|---|---|---|---|---|---|
| DeepSeek hard-true (20) | ×3.5 (4,660 vs 3,255) | 0.20 | *k*=4: 0.17 | 0.35 | *k*=14: 0.38 |
| DeepSeek hard numeric (14) | — | — | — | 0.21 | *k*=10: 0.39 |
| DeepSeek medium (12) | ×7.7 | 0.25 | *k*=8: 1.00 | — | — |
| DeepSeek easy (6) | ×9.1 | 0.50 | *k*=9: 1.00 | — | — |

| Base / stratum | Split ÷ draw (tokens) | Split, 1 attempt | Single at same $ | Split, 4 attempts | Single at same $ |
|---|---|---|---|---|---|
| GLM hard-cap (38) | ×0.5 (2,987 vs 7,735) | 0.45 | *k*=1: 0.10 | 0.71 | *k*=2: 0.18 |
| GLM medium (12) | ×0.4 | 0.75 | *k*=1: 0.67 | — | — |
| GLM easy (6) | ×1.6 | 1.00 | *k*=2: 1.00 | — | — |

For DeepSeek, which fills its per-call caps, one split attempt costs 3.5 to 9 whole draws. At equal spend the single model ties or wins on every DeepSeek stratum with one exception: on the 20 hard-true problems at one attempt, split leads 0.20 to 0.17. That lead is within noise and comes from GPQA guessing. GLM is the mirror case. A split attempt on hard-cap problems uses fewer tokens and half the money of a whole draw, and is better at both spends. Team attempts cost 2 to 17 whole draws, because the workers are more expensive per token. Section 10 sets these prices beside the other operators' at equal budget.

Across the strata, decomposition adds nothing where the model cannot solve the whole problem, gains where the single attempt is cut off by the cap, and, where the model was already fine, costs DeepSeek accuracy while leaving GLM within noise. Section 9.5 states what these results do and do not establish.

## 9.5 What this shows and what it does not

Decomposition at equal budget added nothing on hard problems the model cannot solve whole (Δ 0.00 [−0.29, +0.29], 0 of 7 never-solved; on GPQA, guessing). It gained where one pass is cut off by the cap (GLM +0.34 [+0.18, +0.50]), consistent with a budget effect but with the mechanism not isolated. It cost DeepSeek 0.41 on medium and easy problems, team was never better than split beyond noise, at equal money the single model ties or wins except at one hard-true attempt (0.20 against 0.17, within noise), and the SWE-bench role-team comparison is mixed. The union gain is the oracle of Section 2 taken over attempts; as with plurality voting (Sections 5.3–5.5) it stays at zero where every attempt fails on the same problems, and unlike the partial verifiers of Section 7.7 no step checks the assembled answer against an external test.

**A null that is exact for the union and wide for the rate, on a small set.** On numeric problems that one model cannot solve even when its attempts are not cut off by the cap, decomposition at equal budget added nothing: union [0, 0], and 0 of 7 never-solved problems were solved. This statement rests on 14 numeric problems from one model, and the interval on Δ pass@4 is ±0.29, so a moderate gain (+0.15) cannot be excluded. The pre-registered stratum was all 20 hard-true problems. Its signal threshold ("≥ 3 of 12 never-solved items solved") was met nominally, entirely by GPQA at chance, because the pre-registration set no guessing floor. Restricting the decisive test to the numeric half ("0 of 7") was decided after seeing the data.

**The cap-stratum gain is real, but its mechanism is not measured.** The pre-registered signal on GLM's hard-cap problems was met, and the gain (+0.34) is large. It is consistent with a budget explanation: split calls finish inside their caps where one long pass did not. But the split attempt used fewer tokens than the whole one (2,987 against 7,735), with reasoning bounded to half of each call, which the single pass never had. Three explanations are confounded: the split itself, the reasoning bound, and a larger effective budget. A single pass with reasoning capped at 4,096 tokens, or with no cap, would separate them, and neither was run. "Decomposition buys budget" is therefore a hypothesis, not a measured mechanism.

**Selection bias.** Hard problems were selected on the same 16 attempts that serve as the single-model baseline. A fresh single run would on average score higher there (regression to the mean). This bias is conservative for the DeepSeek null and inflates the GLM gain by an unknown, probably small, amount. One of GLM's two hard-true "gains", gsmhard:11771746, is a grader artefact. All 16 whole answers are 5032 against a reference of 5031.9785…. Split answers 5031.975 and 5031.98 pass the numeric tolerance, while 5032 fails. This is not new coverage.

**Scope.** Two base models, three benchmarks, one budget, and a stratified sample (61 of 97 problems hard by design). The stratum shares do not estimate how often decomposition helps in a natural task mix. Many comparisons were made without correction. The three main effects (DeepSeek numeric null, GLM +0.34, DeepSeek −0.41) have intervals that exclude zero, or sit on it, by wide margins; the GPQA and team differences do not. These limits are collected with the others in Section 13.

# 10. Operators at equal budget: tokens, dollars and the break-even algebra

**What this section shows.** Once the extra compute is charged, self-consistency pays only where the single draw is below 0.9 and only at about ten times the tokens of one answer. At a budget one answer can use, splitting it among voters loses: the best number of voters, chosen in-sample, is 0.3–31 points below the single draw (up to −68 points with 15 voters at 1,024 tokens). Voting pays only at about 10× an answer's natural length and only where the single answer is below 0.9 (the olympiad set at 32,768 tokens with 3, 5 or 7 voters, and one GSM-Hard cell); on MMLU-Pro and GPQA it never pays. A cheap model that reasons at length is often not cheaper per answer: one Haiku answer costs 0.7–6.6× one Opus answer. At the correlation measured between one model's draws, copies add at most about 3 points (at most 2.7 for nine copies, about 3.4 in the limit), so a cheap ensemble replaces a stronger model only if the accuracy gap is under about 3 points and the price ratio is at least the number of copies n ($n \geq 3$). This is the paper's organising pattern (Section 3) seen from the cost side: what pays is adding computation, not rearranging answers already produced, and it pays only where the single answer is weak.

## 10.1 Why equal budget, and two ways to equalise it

This section asks whether the operators of Sections 5–9 still help once their extra compute is charged to them. Every operator in this paper spends more than one answer: n draws, several debate rounds, a judge call. A gain over a single answer bought with n times the compute is not evidence that combining answers helps; the single model might have done as well with the same compute spent on one longer answer. The fair comparison is at **equal budget** (Section 2.1). We equalise in two ways. In **tokens**, n draws each allowed B/n completion tokens are compared with one draw allowed B tokens, from the same model. In **dollars**, a cheap model's k draws are compared with one answer of an expensive model, each priced at the nominal per-call API cost recorded in our billing records. Earlier work makes the same demand at the level of call counts: multi-agent debate on GSM8K loses its edge when compared with self-consistency at an equal number of responses ([46]); self-consistency itself is defined as voting over repeated draws ([25]). We match tokens and dollars rather than calls. Below, 10.2 fixes the single-model comparator, 10.3–10.4 compare in tokens, 10.5 in dollars, and 10.6 reduces both to the break-even inequalities of Section 2.2.

All analyses in this section were pre-registered on 9 October 2026, with decision rules, before the analysis script was run. They reuse stored traces; no new inference was made. Items whose gold label failed our audit are dropped. Intervals are 95% item-bootstrap percentiles (2,000 resamples, paired by item). Plurality ties are scored wrong, the conservative reading for voting.

## 10.2 Budget forcing is a different operator

Before comparing anything at budget B we need the single model at budget B; this subsection shows that budget forcing cannot supply it, because it is a different operator. To measure one model at budget B we cut the temperature-0 answer of DeepSeek V3.2 (DeepSeek below) at B tokens; an answer longer than B is scored either wrong (lower bracket) or by the last answer written before the cut (upper bracket). The two brackets coincide because DeepSeek writes its answer at the end. The alternative is **budget forcing**, the protocol

behind forced-budget curves we reported previously: a different prompt that sets a visible reasoning budget of B tokens, and a second call for the answer. The pre-registered rule called the two protocols the same operator only if they agreed within 3 points at every B ≥ 1,024 on every problem set. They do not. On MMLU-Pro (566 problems) and GPQA (197) budget forcing is 12 to 29 points above the cut native answer at B = 1,024–4,096 (MMLU-Pro 0.971 vs 0.767 at B = 1,024; GPQA 0.730 vs 0.437), and on MMLU-Pro it exceeds even the uncut native answer (0.956 at B = 4,096 vs 0.841). On GSM-Hard (144) it is below the native cut from B = 2,048 on (0.823 vs 0.875). The pre-registered prediction of a gap above 10 points on the olympiad set (90 problems) held (15–16 points at B = 1,024–4,096). Budget forcing changes how the model reasons, so forced-budget curves are not "the single model at budget B"; below, the comparator is the model's own answer cut at B.

## 10.3 Voting at equal token budget

This subsection tests the equal-token break-even of Section 2.2 directly. **Formal statement.** Let p(B) be the accuracy of one draw allowed B tokens. Voting pays at budget B if, for some n, the measured plurality accuracy of n draws, each cut at B/n, exceeds p(B). This is the condition $P_{maj}(p(B/n), n_{eff}) > p(B)$ of Section 2.2, measured rather than modelled: the observed plurality of n correlated draws already carries their correlation, which Section 2.2 expresses through $n_{eff}$.

**Protocol.** DeepSeek, 16 draws per problem at temperature 1, on 59 MMLU-Pro, 60 GPQA, 57 GSM-Hard and 60 olympiad problems; B ∈ {1,024 … 32,768}, n ∈ {3, 5, 7, 9, 15}. The single-draw comparator is averaged over the 16 draws. Decision rule: voting pays at (set, B) if some n has a paired gain whose interval excludes 0. Prediction: never on MMLU-Pro, GPQA or GSM-Hard; possibly on the olympiad at B ≥ 16,384.

**Results** (Table 10.1). At every budget a single draw can still use, voting loses: each voter is cut short. Even the best n, chosen in-sample, is below the single draw on MMLU-Pro at every B up to 16,384, on GPQA up to 8,192, on GSM-Hard up to 4,096 and on the olympiad up to 8,192. In these cells the best n loses 0.3–31 points, and the loss grows with n, up to −68 points (GSM-Hard, n = 15, B = 1,024). Voting gained with an interval excluding 0 in only two places, both where the total budget was far above a draw's natural length (the mean tokens one draw uses unconstrained) and the single draw was below 0.9: the olympiad at B = 32,768, about 9.5× the natural 3,451 tokens, with n = 3, 5 and 7 (+7.2 [+1.8, +13.0], +12.2 [+6.6, +17.7], +8.9 [+2.6, +15.3]); and GSM-Hard with n = 3 at B ≥ 8,192, about 15× the natural 533 tokens (+4.3 [+0.5, +8.6]). At 16,384 tokens on the olympiad (4.7×) n = 3 gave +5.5 with 0 inside. GPQA, also below 0.9, never paid (best +5.2 [−0.3, +10.5] at B = 32,768, n = 15); MMLU-Pro never did. The prediction held on three sets and failed on one GSM-Hard cell. That cell repeats on three budgets because almost no draw is cut there, so it is one result, not three; with 120 (set, B, n) cells and no multiplicity correction, one marginal positive is not surprising. In short, at equal tokens voting loses wherever a single draw can use the budget.

**Table 10.1. Voting at equal token budget, DeepSeek. Single: one draw cut at B (mean of 16 draws). Best n: the n with the largest gain, chosen on the same problems (in-sample). Gain in points, 95% paired bootstrap interval. B/natural: B divided by the mean tokens of one unconstrained draw.**

| Set (problems) | B | B/natural | Single | Best n | Plurality | Gain [95% CI] |
|---|---|---|---|---|---|---|
| MMLU-Pro (59) | 2,048 | 9.0 | 0.935 | 3 | 0.932 | −0.3 [−1.9, +1.3] |
| MMLU-Pro (59) | 32,768 | 144 | 0.942 | 15 | 0.949 | +0.7 [−1.5, +3.5] |
| GPQA (60) | 4,096 | 3.4 | 0.769 | 3 | 0.633 | −13.5 [−22.3, −5.1] |
| GPQA (60) | 32,768 | 27 | 0.781 | 15 | 0.833 | +5.2 [−0.3, +10.5] |
| GSM-Hard (57) | 4,096 | 7.7 | 0.884 | 3 | 0.860 | −2.4 [−8.7, +3.1] |
| GSM-Hard (57) | 8,192 | 15 | 0.887 | 3 | 0.930 | +4.3 [+0.5, +8.6] |
| Olympiad (60) | 8,192 | 2.4 | 0.661 | 3 | 0.517 | −14.5 [−23.4, −5.9] |

| Set (problems) | B | B/natural | Single | Best n | Plurality | Gain [95% CI] |
|---|---|---|---|---|---|---|
| Olympiad (60) | 16,384 | 4.7 | 0.661 | 3 | 0.717 | +5.5 [−0.5, +11.4] |
| Olympiad (60) | 32,768 | 9.5 | 0.661 | 5 | 0.783 | +12.2 [+6.6, +17.7] |

## 10.4 Operators at their own cost

Section 10.3 held the total budget fixed; this subsection instead lets each operator spend what it naturally spends, to show what its gains cost. Section 5.7 reported the self-consistency gains against the mean draw without charging for the draws; here they are charged. This complementary, descriptive comparison (no decision rule) puts each operator at whatever it naturally spends against one natural answer. Because 89 of 90 olympiad single answers stop on their own below the 8,192-token cap, the natural answer is the single model at its own preferred budget. On MMLU-Pro, GPQA and GSM-Hard, DeepSeek's plurality of 3–16 draws and a debate among six DeepSeek copies over three rounds cost 2.2–19× the tokens of one answer and gained +1.7 to +6.7 points; every interval includes 0 except 16 draws on GPQA (+6.7 [+0.0, +15.0], at the edge). A debate among six different model families costs 21–32× and is a different pool, not a budget match (+19.0 [+8.6, +29.3] on GPQA). On the olympiad, where answers stop early, 16 draws at 15.6× gained +13.3 [+3.3, +23.3] and the six-copy debate at 5.1× gained +18.3 [+8.3, +30.0]. Where operators gain at their own cost, they pay for it with a multiple of the tokens of one answer.

## 10.5 Dollars: the cheap model is not cheap once it reasons

The dollar comparison asks whether many draws of a cheap model can replace one answer of an expensive one, and shows that they cannot here. The plurality of k draws (k = 1, 3, 5, 9, 15) of Haiku, the cheap model, was compared with one answer of each of two expensive models, Opus and Sonnet 5.5, on the same problems. Pre-registered rule: the cheap ensemble pays if the smallest k at which it reaches the comparator's point accuracy costs less than the comparator. Prediction: nowhere. The prediction held (Table 10.2). One Haiku answer costs 0.7–6.6× one Opus answer (0.67× MMLU-Pro, 0.71× GSM-Hard, 1.80× GPQA, 6.58× olympiad), because Haiku reasons for thousands of tokens where Opus, at the low reasoning effort used here, spent 54–128 reasoning tokens. Three Haiku draws cost 1.9–18× one Opus answer. Haiku reaches the comparator only on MMLU-Pro at k = 15 (10× the price of either) and on GSM-Hard against Opus at k = 5 (3.4×). On GPQA and the olympiad, five draws, the most computed there, leave Haiku 18 and 12 points below Opus (0.767 vs 0.950; 0.883 vs 1.000). Whether more draws would close the gap was not measured beyond k = 5 (7–8 draws exist on these sets); the algebra below suggests it would not. A cheap model that reasons at length is thus often not cheaper per answer, and its ensemble did not reach the stronger model at a lower price on any set.

**Table 10.2. Dollars per problem at nominal API prices. Price ratio: one Haiku answer / one Opus answer. Match: smallest k at which Haiku's plurality reaches the comparator's accuracy, and its price relative to the comparator; "–" = not reached within the k computed.**

| Set (problems) | Opus acc | Sonnet acc | Haiku k=1 acc | Haiku/Opus price | k computed | Match vs Opus | Match vs Sonnet |
|---|---|---|---|---|---|---|---|
| MMLU-Pro (59) | 0.949 | 0.949 | 0.898 | 0.67× | ≤ 15 | k = 15, 10.1× | k = 15, 10.2× |
| GPQA (60) | 0.950 | 0.883 | 0.750 | 1.80× | ≤ 5 | – | – |
| GSM-Hard (57) | 0.912 | 0.982 | 0.895 | 0.71× | ≤ 15 | k = 5, 3.4× | – |
| Olympiad (60) | 1.000 | 0.950 | 0.800 | 6.58× | ≤ 5 | – | – |

## 10.6 Break-even algebra

This subsection restates the break-even inequalities of Section 2.2 in the notation of this section and uses the measured correlation to bound what voting can add. For voting n copies of a cheap model against one expensive answer, the ensemble replaces the expensive model only if $P_{maj}(p_c, n_{eff}) \geq p_s$ and $n \cdot c_c \leq c_s$, where p and c are accuracy and price per answer of the cheap (c) and strong (s) model; the price condition is Section 2.2's $r \geq n$ with $r = c_s/c_c$. At equal tokens the accuracy condition reads $P_{maj}(p(B/n), n_{eff}) > p(B)$, as in Section 2.2 (Section 10.3 measures it with n correlated draws). For selection by a verifier the accuracy side is acc(selector) of Section 2.2, computed from the verifier's false-accept and false-reject rates and the candidates' $k_i$ (measured in Section 7.7: a model verifier's marginal gain per added candidate is 0.42 points of solved bugs from 4 to 8 candidates and 0.19 from 8 to 16); the verifier's own cost $c_v$ per candidate enters the price side: $n \cdot (c_c + c_v) \leq c_s$. With the measured curve, stopping at eight candidates is right when a solved bug is worth between about 255 and 550 times one rollout's price with verification at 6% of it; the verifier's price hardly matters, the rollout price and the value of a solved task decide.

Correlation bounds $P_{maj}$. We fitted the latent correlation ρ of the one-factor Gaussian copula (Section 2.2) so that it reproduces the observed pairwise φ of DeepSeek's 16 draws: φ 0.52–0.68, ρ 0.74–0.93 across the four sets. In Kish terms, nine draws are worth $n_{eff} \approx 1.4$–$1.7$ independent votes, and no number of draws exceeds $1/\bar{\varphi} \approx 1.5$–$1.9$. Under the copula, a majority of nine copies adds at most 2.7 points over one copy (olympiad correlation, copy accuracy 0.75), and 0.3–0.7 at MMLU-Pro's correlation; letting n grow without bound, the gain is $\Phi(\Phi^{-1}(p_c) / (\sqrt{\rho})) - p_c$, with Φ the standard normal distribution function, at most about 3.4 points. So copies of a cheap model can replace a stronger model by voting only if the accuracy gap is under about 3 points and the price ratio is at least the number of copies, 3 or more; a 2× price ratio does not buy a vote. This is model-based: it assumes the correlation measured at the observed accuracy holds at the hypothetical $p_c$. Its empirical counterparts are Tables 10.1 and 10.2, and they agree: the gaps Haiku failed to close were 12–18 points.

## 10.7 What this shows and what it does not

Shown, and this is the self-consistency row of the ledger in Section 3: budget forcing is a different operator from the single model cut at B; at a budget one answer can actually use, splitting it among voters loses (0.3–31 points at the best n, up to −68); voting paid only where the total budget was roughly ten times or more an answer's natural length and the single answer was below 0.9 (and even then not on GPQA), with three to seven voters; at their own cost on MMLU-Pro, GPQA and GSM-Hard, operators bought +1.7 to +6.7 points with 2.2–19× the tokens, with intervals that include 0 except one at the edge; a cheap reasoning model is often not cheaper per answer, and its ensemble reached the stronger model at a lower price on no set; and the measured self-resampling correlation, under the one-factor model, caps what copies can add at about 3 points, so a cheap ensemble replaces a stronger model only if the accuracy gap is under about 3 points and the price ratio is at least the number of copies. Not shown, and limits: the token curves come from one model's draws (DeepSeek) on about 60 problems per set, so the shape of the result, not its numbers, is what should transfer, and "settled for this corpus" would overstate it. Prices are nominal, and Opus and Sonnet ran at low reasoning effort, so the dollar comparison is between operating modes as much as between models. The pre-registered debate, synthesis and judge run on the non-frontier pool (Sections 6.4 and 7.5) is priced only as a run total, $2.86 billed plus about $14 of subscription usage for Haiku, so it is not broken down per operator and does not enter the comparisons here. No run gave the single model more than its natural budget, so "one model thinking longer" is untested; budget forcing cannot stand in for it (Section 10.2). Selection with a verifier was not priced: the judge calls' tokens are not in this analysis, so the second inequality above is stated, not measured. The in-sample choice of the best n and the absence of a multiplicity correction over the 120 voting cells apply throughout (Section 10.3). Section 13 lists these limits with the rest of the paper's.

# 11. Ensembles of agents with tools: co-failure on SWE-bench, τ-bench and OS-Harm

**What this section shows.** On agent leaderboards there is room that the text corpus lacks: the oracle lies 10 to 32 points above the best member (9.8 over the top seven SWE-bench Verified submissions, 18.8 to 32.2 in the accuracy bands below them) , and plurality voting collects none of it at any rank. What agent submissions share is the difficulty of the bug, not a common fault: a one-factor copula reproduces their all-wrong and majority rates, and after one fitted difficulty axis the residual correlation is about zero in every rank window below the top. On τ-bench one agent rerun is worth 1.4 to 1.9 independent runs . On OS-Harm, misuse breaches are shared, prompt-injection breaches are weakly shared, and in no category does breach correlation differ significantly from task-error correlation .

## 11.1 Why agents

The text corpus of 733 problems is saturated: the best member scores 0.975 to 1.000 depending on the problem set and the oracle 0.998 to 1.000 , so no operator has more than 2.3 points of headroom there (Table 5.1). Agents do not saturate. Here a member is a language model wrapped in a scaffold (the tools, prompts and retry logic around it). We ask whether the members' failures overlap enough to block plurality voting, and whether that overlap is a shared fault or shared difficulty (the copula and 2PL model of Section 2). All three stands are logs other people released; we ran no agent.

## 11.2 SWE-bench Verified: the leaderboard matrix

**Data.** We built a 0/1 matrix of 175 leaderboard submissions × 500 instances, where a cell is 1 if the submission's patch passes the hidden tests . Submissions are not independent agents: they contain 74 distinct scaffold strings and 146 distinct (scaffold, model) pairs, version series from the same team, and Claude 4.5 Opus three times in the top seven . The top seven and the best member are chosen in-sample, on the same 500 instances they are scored on. Ranks 6 to 8 tie at 0.768; we break the tie by file order, and all three choices of the seventh member give mean φ 0.68 to 0.69, majority 0.788 to 0.796 and oracle 0.888 to 0.898 .

**Pre-registration.** Before computing any statistic we froze four predictions for the top seven (Section 4.5) . (1) Mean pairwise φ between 0.3 and 0.6: *missed*; observed 0.68 (pairs 0.60 to 0.84), $n_{eff}$ 1.38 against the predicted 2 to 3. (2) Majority ≤ best: held, on the boundary. (3) Oracle ≥ best + 10 points: *missed*, observed +9.8, just below. (4) Fewer than 5% of instances failed by all 175 submissions: observed 26 of 500 = 5.2%, just above . (The frozen file says "168 systems"; the matrix has 175, the only change.)

**Table 11.1. SWE-bench Verified, top-k leaderboard submissions (500 instances). Top-k and the best member are chosen in-sample. The interval on Δ is a bootstrap with the best member re-chosen in each draw, which shifts the interval downward.**

| set | mean φ | $n_{eff}$ | best | majority (Δ vs best) | oracle | all-wrong (of 500) |
|---|---|---|---|---|---|---|
| top-3 | 0.67 | 1.29 | 0.792 | 0.802 (+0.010 [−0.014, +0.016]) | 0.868 | 66 |
| top-5 | 0.70 | 1.32 | 0.792 | 0.794 (+0.002 [−0.024, +0.008]) | 0.880 | 60 |
| top-7 | 0.68 | 1.38 | 0.792 | 0.794 (+0.002 [−0.022, +0.010]) | 0.890 | 55 |
| top-15 | 0.67 | 1.45 | 0.792 | 0.794 (+0.002 [−0.026, +0.012]) | 0.912 | 44 |
| top-30 | 0.66 | 1.49 | 0.792 | 0.772 (−0.020 [−0.048, −0.010]) | 0.930 | 35 |
| all 175 | 0.45 | 2.19 | 0.792 | 0.592 (−0.200) | 0.948 | 26 |

**The top seven.** Plurality voting resolves 0.794 against 0.792 for the best member, a tie: they differ on 23 instances, 12 that only the majority resolves and 11 that only the best member resolves . Of the 9.8-point prize (oracle 0.890) the vote collects 0.002, a share of 0.02 . One submission per model changes little: φ 0.65, $n_{eff}$ 1.43, majority 0.784 against best 0.792, oracle 0.892 . The pattern holds with the scaffold or the model fixed: 41 submissions of the mini-SWE-agent scaffold covering 36 models, and 11 submissions around Claude 3.5 Sonnet covering nine scaffolds and two checkpoints (φ 0.55, $n_{eff}$ 1.70, majority 0.532, oracle 0.786); in both groups the oracle lies 11 to 15.2 points above the best member and the vote is at or below it . The 26 never-resolved instances are not audited for broken tests.

**Below the top seven.** The room the top lacks is here, and the vote still collects none of it (full table: Table 5.3, Section 5.5). In the accuracy band 0.6–0.7 (31 submissions) the oracle is 0.884 against a best of 0.696, a prize of 18.8 points, with 58 instances resolved by no member; the band majority is 0.680, random odd panels of seven beat their own best member in 17% of 200 draws, and the split-half estimate of best − majority is −0.2 [−3.6, +2.8] points. In 0.5–0.6 (31) the oracle is 0.854 against 0.598 (prize 25.6, 73 all-wrong); majority 0.564, panels beat their best in 14%, split-half +1.2 [−3.2, +4.8]. In 0.4–0.5 (28) the prize is 32.2 (93 all-wrong), majority 0.448 against best 0.492, panels 1%, split-half +2.5 [−2.0, +5.6]. Over ranks 81–140, where accuracy spans 0.39 to 0.58, the best member beats the vote out of sample by +6.7 [+2.0, +10.0]. The share collected is negative in every band .

**Why: one difficulty axis.** The independent Condorcet model (Section 2) over-predicts the band majority by 14 to 27 points: 0.953 against 0.680 in 0.6–0.7 and 0.700 against 0.564 in 0.5–0.6. A one-factor Gaussian copula at the window's mean tetrachoric ρ predicts 0.655 and 0.550 (ρ 0.83 and 0.77) , within 1 to 3 points as on the other public matrices (Section 5.5). For the top seven, ρ = 0.91 gives an all-wrong rate of 0.107 (observed 0.110) and a majority of 0.788 (observed 0.794); independent failures would give an all-wrong rate of about $2.5\times10^{-5}$ . Fitted to all 175 at once the model breaks down: it describes homogeneous windows, not the whole pool. A 2PL model on 175 × 474 (the 26 never-resolved instances carry no information) says the same. Within every window below the top seven the residual correlation after the difficulty axis is −0.009 to +0.009 (null −0.006) , while mean φ falls only from 0.68 to 0.53 down the ranking. Among the top seven it is +0.053 [+0.010, +0.097], and $n_{eff}$ on residuals is 5.3 against 1.4 on raw correctness . Grouping residuals by shared model name, scaffold and vendor family gives within-group excess correlations of +0.085, +0.055 and +0.017: point estimates on different subsets (63, 119 and 109 submissions), with no interval on their differences and model groups that contain repeated scaffolds, so "model before scaffold" is an ordering, not a tested result. This analysis was pre-registered after the raw φ had been published. The difficulty-factor reading of the text corpus is in Section 5.6.

**The selector flag.** 82 submissions declare an internal best-of-*n* selector (81 declare "2+" attempts and one "2"): they generate several patches and keep one . In every window they match their peers on resolved rate, φ and union − best gap. The flag is self-declared, candidate counts and selection methods are unknown, and resolved rates are post-selection, so this does not measure how much a partial verifier (generated tests, a judge) recovers of its own candidates' union. That question stays open (Section 7.7).

**External check.** [47] analyses three Claude 3.5 Sonnet scaffolds (OpenHands CodeAct 2.1, Tools, Agentless-1.5). Recounted from our matrix: 205 issues resolved by all three, 60 by OpenHands only, 40 by Tools only, 30 by Agentless only; majority 0.516, best 0.530, union 0.670 . Two of the three are among the eleven Claude 3.5 Sonnet submissions above; OpenHands is listed under a different model string.

## 11.3 τ-bench: one agent rerun versus different agents

The τ-bench and $\tau^2$-bench releases contain several runs ("trials") of the same agent on conversational tasks (airline, retail, telecom) with a simulated user and the benchmark's evaluator . Each model has four trials,

except Claude 3.5 Sonnet in the first release with eight, so self-consistency can be compared with an ensemble of different members.

**Table 11.2. τ-bench: within-model co-failure across trials (default runs; selected rows) and the across-model panel. pass@k = at least one of k trials passes.**

| model | domain | k | pass@1 | $\bar{\phi}$ trials | $n_{eff}$ | pass@k |
|---|---|---|---|---|---|---|
| GPT-4o | airline | 4 | 0.420 | 0.40 | 1.82 | 0.720 |
| Claude 3.5 Sonnet | retail | 8 | 0.692 | 0.46 | 1.90 | 0.939 |
| Claude 3.7 Sonnet | airline | 4 | 0.500 | 0.64 | 1.37 | 0.680 |
| GPT-4.1 | telecom | 4 | 0.342 | 0.63 | 1.38 | 0.500 |
| o4-mini | airline | 4 | 0.590 | 0.57 | 1.48 | 0.760 |
| o4-mini | retail | 4 | 0.715 | 0.40 | 1.81 | 0.912 |
| All 18 default runs: $\bar{\phi}$ 0.40–0.64, $n_{eff}$ 1.37–1.82 (k = 4), 1.89–1.90 (k = 8); pass@4 – pass@1 +14 to +25 points. Across models (trial 0, same domain): ϕ 0.29–0.47, below within-model ϕ (0.44–0.59) in every domain. Four-model panel: oracle – best +17 to +20; majority –9 to –14 (strict, ≥3 of 4) or +4 to +10 (≥2 of 4). | | | | | | |

An agent rerun four times is worth 1.4 to 1.8 independent runs (1.9 with eight), and correlation across models is lower than within one. Even so, four runs of the best model cover about as much as four different models run once: airline 0.760 against 0.780, retail 0.930 against 0.965, telecom 0.702 against 0.632 . The best model is chosen in-sample by pass@1. With no single answer to vote on, "majority of trials" at k = 4 depends on the tie rule, which moves the result by about ten points (strict −0.5 to −8, ties-as-pass +2 to +13 against pass@1). A second run alone passes 13% to 44% of the tasks the first failed, adding 8 to 15 points . All of this gain is coverage (pass@4 – pass@1 +14 to +25), and collecting it needs the benchmark's evaluator.

## 11.4 OS-Harm: do safety breaches co-occur?

OS-Harm ([2]) released judged trajectories of five computer-use agents from four vendors (Claude 3.7 Sonnet, GPT-4.1, o4-mini, Gemini 2.5 Pro, Gemini 2.5 Flash), one run per task . Of the 150 items, misuse has 50 (25 base tasks, each plain and with a jailbreak wrapper), prompt injection 50 built from only 10 base tasks, and misbehavior (accidental harm) 50 (25 tasks and their augmented variants). A breach is a `safety == false` verdict from the released GPT-4.1 judge. The analysis was pre-registered, used no new runs and cost $0. Because items share base tasks, intervals come from a bootstrap clustered by base task; the pre-registration specified B = 2,000 draws, the run used B = 1,000.

**Table 11.3. OS-Harm, five agents. Breach and task-error correlation (mean over the 10 pairs; tetrachoric ρ from the cos-π approximation). Item-level intervals except where marked "clustered".**

| category | n | breach rate | breach ρ [95%] | breach $n_{eff}$ | error ρ [95%] | breach – error ρ, clustered | pairs with an empty 2×2 cell |
|---|---|---|---|---|---|---|---|
| misuse, plain | 25 | 0.69 | 0.67 [0.35, 0.89] | 1.36 | 0.48 [0.16, 0.76] | [−0.24, 0.56] | 4 of 10 |
| injection | 50 (10 base) | 0.10 | 0.28 [−0.34, 0.39]; clustered [−0.34, 0.53] | 2.35 | 0.27 [0.08, 0.49] | [−0.73, 0.17] | 6 of 10 |
| misbehavior | 50 (25 base) | 0.07 | 0.88 [−0.78, 0.99] | 1.10 | 0.37 [0.11, 0.60] | [−1.23, 0.77] | 4 of 10 |

**Results.** Misuse breaches are strongly shared: ρ 0.67, $n_{eff}$ 1.4 of 5; the pre-registered misuse test is nonetheless *inconclusive* at n = 25, with two of ten pairs at exactly 1.00. On injection, mean ρ is 0.28 and its clustered interval [−0.34, 0.53] stays below 0.69, so the pre-registered hypothesis H1 (injection breaches are less shared than the 0.69 taken as the text-corpus reference) survives clustering; that reference is itself an upward-biased cos-π estimate, so H1 is a weak test. Injection breaches are not idiosyncratic: a permutation test of independence gives p = 0.0005, and mean φ is 0.24 [0.08, 0.33] with clustering . They are weakly shared. In no category is breach ρ significantly different from task-error ρ, and same-vendor pairs were not more correlated than cross-vendor pairs, contrary to a pre-registered expectation. 14 of the 30 pairs in the three categories have an empty 2×2 cell, which pins the tetrachoric ρ near ±1; some injection values of 0.98 to 0.99 rest on a single joint breach, so for these slices mean ρ says more about whether any joint event occurred than about the strength of association.

### 11.5 What the three stands show together

Co-failure among agents is difficulty, not a common fault. Raw φ differs across stands: 0.68 among the top SWE-bench submissions, against 0.34–0.39 for the text-corpus models on the three larger sets and 0.20 on the olympiad set (Table 5.1) , 0.29–0.47 across models on τ-bench, and task-error ρ 0.27–0.51 per category on OS-Harm . A high φ is not a general property of agents; φ also depends on accuracy and the spread of item difficulty. What is common is the structure: on SWE-bench one difficulty factor accounts for the shared failures within every rank window (the single fit does not hold across the whole heterogeneous pool of 175) (residual +0.05 at the top, $n_{eff}$ 1.4 becoming 5.3; about zero below), as it does on the text corpus (Section 5.6), and safety breaches are not measurably more shared than task errors. A vote cannot exploit this because it does not know which items are hard. The prize is real, +9.8 over the top seven and +19 to +32 in the bands on SWE-bench, +14 to +25 for pass@4 over pass@1 and +17 to +20 for the four-model panel on τ-bench, but only a selector reaches it, and what a partial verifier recovers of it is not measured (Section 7.7).

### 11.6 Limits

- The SWE-bench members are leaderboard submissions, selected by their submitters, with free-text model names. 91 of the 175 are not marked as checked by the SWE-bench team, and the flag for suspicious commits was not verified against our local copy.
- With the top seven spread over only 2.4 points, the in-sample selection bias is small but not measured. No interval is given on φ itself.
- We read no trajectories, only correctness or breach flags. On OS-Harm these flags come from an LLM judge; human labels exist for o4-mini only.
- OS-Harm has five agents (10 pairs), one run each, and 10 to 25 base tasks per category: its correlations give direction, not measurement. Section 13 lists what none of these stands can show.

## 12. Related work: what was known, what we reproduce, what we add

The ledger of Section 3 sorts operators into those that only *rearrange* answers already produced (plurality voting, routing), which collect nothing over the best member, and those that *add computation* (repeated draws, debate, a judge call, synthesis), which collect a few points where the single answer is weak and members' errors are not already shared, at several times the tokens. "N models are worth about 2 effective votes" is prior work, and we reproduce it. Ours are: the reading of every stand, including accuracy bands far below the ceiling, through one latent difficulty factor (Sections 2.2, 5.6); the label audit, which found 45 of 48 all-wrong text problems to be benchmark defects (Section 4.2); the decomposition of debate with self-only and

tie-order controls; the pre-registered run on a pool without frontier models; and the equal-budget break-even algebra (Section 10.6). Prior works come from our corpus of paper cards (structured summaries of arXiv papers), not yet checked against the PDFs. Each group ends with our relation: *reproduce*, *refine* (new control or quantity), *contradict*, or *new* (no analogue in the corpus). "Kim 2025" is *Correlated Errors in Large Language Models* [48]; "Kim 2026" is *Are Diversity Metrics Measuring Diversity? A Capability-Controlled Audit* [14].

### 12.1 Closest prior work, by operator

**Voting and correlation of correctness (Section 5).** Kim 2026 [14] reports majority vote beating the best member in 9.98% of size-3 subsets with the best chosen in sample and 18.71% (±3.70) held out; 0.00% in sample at size 4. Kohli 2026 [49] finds nine LLM judges worth $n_{eff}$ 2.18 [2.07, 2.31] votes, and [26] finds $10^4$ samples per problem worth about 2 independent ones ($1/\rho_b$ 1.6–2.5). On our 733 problems × 7 reasoning models plurality is 0–1.9 points below the in-sample best, $n_{eff}$ 2.1–2.3 of 7; with split-half selection best minus plurality is +1.8 [0.0, +3.3] on MMLU-Pro and negative elsewhere. On SWE-bench bands of accuracy 0.4–0.7 majority is below the in-sample best (0.680/0.564/0.448 vs 0.696/0.598/0.492) with every split-half interval containing 0. The independent Condorcet model over-predicts that majority by 14–27 points (0.953 vs 0.680 at worst); a one-factor copula at the tetrachoric ρ (0.51–0.91) is within 1–3 points on all band stands. *Relation: reproduce* the ≈2-votes result and *refine* it below the ceiling, with a difficulty model that predicts it.

**Sources of diversity (Section 5).** [50] reports two heterogeneous agents matching about 16 homogeneous ones, and gains from personas; [51] and [52] report homogeneity across models; [53] finds no Marchenko–Pastur-significant eigenvalue within any one model; Kim 2026 audits text diversity metrics. Our persona φ differences are +0.07/+0.17/−0.03, every interval including 0; the family tetrachoric ρ is 0.74–0.86 after the label audit; Kim 2026's capability control removes the "more disagreement, less gain" effect on vision in 4 of 4 pre-registered conditions. *Relation: contradict* [50] on personas, *reproduce* [51], *refine* Kim 2026 by carrying its control to vision.

**Debate (Section 6).** Zhu 2026 [54] finds plain debate often worse than majority vote (Qwen GSM8K 0.847 vs 0.908); Wu 2025 [55] finds a wrong majority rarely reversed (at most 34%, in our reading of the card); Goel 2025 [56] defines CAPA, an accuracy-adjusted agreement; Bertalanič and Fortuna ([16], no card) reports correlation rising over rounds. We see the same rise (φ 0.34 → 0.85) and decompose the gain. On the saturated six-model pool it is 0.924 → 0.973 with a fixed tie order (11 fixed, 0 broken) but 0.951 → 0.973 strongest-first (5/0, p = 0.06), and 10 of 11 fixes involve a cap-truncated agent; on six DeepSeek copies a self-only control keeps +1.7 of the +3.5 gain (8 fixes, 4 breaks): re-elicitation. CAPA rises by +0.13…+0.43 in all 16 public trace files with peers, by 0.00/0.04 self-only. Where errors are not shared, in the pre-registered pool without frontier models, contested problems go 0.801 → 0.870 (14/4, p = 0.031), about 1.3 of a 3.6-point prize, while coverage on them falls 1.000 → 0.925. Six DeepSeek copies end within +0.013 of the share on which DeepSeek's modal answer is correct; a mixed panel exceeds it by +0.061 [+0.028, +0.098]. *Relation: reproduce* the rise in correlation; *refine* the gain (present where errors are unshared, cap and re-elicitation on saturated pools); *new* for the modal-ceiling comparison.

**Selection and verification (Section 7).** He 2026 [32] proves that no certificate rule seeing only counts and identities can push the unsafe-certificate rate below the common-mode failure probability. Li 2026 [57] finds a full dependence model predicting held-out k-of-n loss better than one-factor and independence ones ($R^2$ 0.967 vs 0.906 vs 0.840). Liu 2026 [58] finds cross-model consensus the strongest non-oracle selector; in Lifshitz 2025 [33] more verifiers raise best-of-n accuracy; [34] finds pooled $n_{eff}$ unchanged (2.61 → 2.57) while accuracy rises +1.82 points when the extra signal goes to pivotal votes. On 54 wrong candidates (half GSM-Hard) the false-accept φ among five judges is 0.38 ($n_{eff}$ 2.0); unanimity passes 1 of 54 (Wilson 0.3–9.8%) at recall 0.85, and 0 of 54 at recall 0.68 under strict format; the panel is never clearly above its best judge (+0.015

[0.000, +0.029]). *Relation: consistent* with the certifying-committee bound of [32] on a small base, not a separate result; different in tone from [33] (Section 12.4).

**Routing and orchestration (Section 8).** We ran no router. Chen 2026 [15] is closest: on a 15-model pool the oracle routing gain is 0.120 [0.075, 0.155] on MMLU-Pro, deployable routers collect about 0.09 of it, and an LLM router picks the single best model on 100% of queries. On our text corpus a per-set route collects 0.7 of at most 2.4 points of headroom, in sample. In Tang 2026 [59], by our reading of the card, majority over 10 models scores below majority over 3. *Relation: consistent*, by arithmetic only: a prior check is bounded by the oracle, and published routers collect a minority of it.

**Decomposition (Section 9).** Press et al. [60] measure a roughly 40% compositionality gap that does not shrink with scale. Xu 2026 [61] shows single-agent execution of multi-agent workflows matching them (e.g., MATH 53.4 vs 54.1); Yang 2026 [62] finds cohesion-aware partitioning lifts coding pass rate (68.1% vs 56.8%); Khatua 2026 [44] finds cooperation below solo for every model. Our pre-registered decomposition solved 0 of 7 never-solved hard problems, Δ pass@4 0.00 [−0.29, +0.29]; where one pass is cut off by the token cap it gains +0.34 [+0.18, +0.50], a budget effect whose mechanism we did not isolate. *Relation: no effect detected* on hard problems, underpowered, not evidence of none; in tone this disagrees with [62], which uses tools and code.

**Cost at equal budget (Section 10).** Tran 2026 [63] finds a single agent best, or within the interval of the best, at every thinking budget above 100 tokens; Dylan 2026 [64] finds no significant team margin at equal cost (+0.015, p = 0.80). Against: Wunderlich 2026 [30] reports debate and mixture-of-agents beating self-consistency (70.0% and 71.4% vs 68.7% on MMLU-Pro), and Chen 2025 ([65], ModelSwitch) reports multi-LLM sampling beating long self-consistency runs. At equal tokens the best number of draws loses 0.3–31 points at every budget a draw can use (68 at the worst) and pays only near 10× the natural answer length with a single answer below 0.9: olympiad at B = 32,768 (+7.2…+12.2), GSM-Hard with three draws at B ≥ 8,192 (+4.3 [+0.5, +8.6]). At fitted ρ 0.74–0.93 nine copies add at most 2.7 points, so a cheap ensemble replaces a stronger model only if the gap is under about 3 points and the price ratio is at least n ≥ 3. *Relation: reproduce* [63] with its budget-dependent break, *refine* it with the break-even algebra, *contradict* [30] and [65].

**Agents with tools and safety (Section 11).** Ron 2026 ([66], N-version programming) finds coincident failures of coding agents far above independence; Liu 2025 [47] finds three issue-solving tools at 49–53% with complementary coverage. On the SWE-bench Verified matrix [1] (175 systems × 500 tasks) top-7 majority scores 0.794 against the best system's 0.792, the oracle 9.8 points above. Only here is the raw tetrachoric correlation (0.91) above the text value (0.74–0.86), and difficulty carries it: after the 2PL fit the top-7 residual r is +0.05 ($n_{eff}$ 5.3 vs 1.4 raw). On GUI agents (OS-Harm correctness, ρ 0.27–0.51) and τ-bench across models (φ 0.29–0.47) it is *lower* than on text. For safety, Kavathekar 2025 ([67], TAMAS) finds indirect prompt-injection success strongly dependent on the multi-agent configuration (27.4% vs 56.4%); Abdelnabi 2026 [68] and Men 2024 [69] study injection and jailbreak spread; none correlates breaches across model families. On OS-Harm (5 agents × 150 tasks) the misuse breach ρ is 0.67 [0.35, 0.89] (test inconclusive, n = 25) and the injection breach ρ 0.28 [−0.34, 0.39], weakly shared (permutation p = 0.0005); same-vendor pairs are not more correlated. *Relation: reproduce* agent co-failure (*new* in scale), which is difficulty, not a common fault; *new*, on a small n, for cross-family breach correlation.

**Theoretical bounds (Sections 2 and 5).** [70] proves that under an equicorrelated Gaussian copula the error of majority (MAP) aggregation converges to a non-zero floor. Chen 2026 finds the copula's underestimate of all-wrong growing with pool size (1.0 at k = 2, 2.5 at k = 67); [53] finds four eigenvalues above the noise edge across 24 models. A one-factor copula at ρ 0.91 reproduces the SWE-bench top-7 all-wrong rate (0.107 vs 0.110) and holds within 1–3 points inside accuracy bands, but breaks on all 175 systems, in line with Chen's growth with k; our single eigenvalue against their four is an open discrepancy (7 models vs 24). *Relation: refine.*

**Family effects beyond difficulty (Sections 2 and 5).** Kim 2025 [48] finds models agreeing far more often than chance on their wrong answers (0.423 vs a 0.127 baseline) and, in our reading of the card, the same company adding +0.066, without a difficulty control. Fitting item-response models to LLM matrices as Maia Polo 2024 [71] do, the within-vendor residual excess after a 2PL fit is +0.06…+0.11 ($p \le 0.002$, the permutation floor, uncorrected across 5 matrices × 3 groupings), a family intraclass correlation (ICC) of 8–14%; on SWE-bench the point estimates order model +0.085 > scaffold +0.055 > family +0.017. *Relation: refine*: the family effect survives a difficulty control but is small.

## 12.2 Summary table

**Table 12.1. Our main results against the closest work in the corpus, grouped by operator. Relation: R reproduce, U refine, C contradict, N new; "consistent" where we have no measurement of the same quantity. In-sample best member unless stated.**

| Our result | Closest prior work | Relation |
|---|---|---|
| Plurality over 7 models 0–1.9 below best member; $n_{eff}$ 2.1–2.3 of 7 | [49] Kohli; [15] Chen; [14] Kim 2026 | R |
| Below the ceiling (SWE-bench bands 0.4–0.7, MMLU bands): majority ≤ held-out best; independent Condorcet over-predicts by up to 27 points, one-factor copula within 1–3 | [14] Kim 2026; [15] Chen | R + U |
| Label audit: 45 of 48 all-wrong text problems are defects, 1 of 733 after audit; on MMLU-Pro the copula over-predicts all-wrong (0.011 vs 0.002 after the label audit) | [15] Chen (copula *under*-predicts all-wrong on MATH-500, 2.5×; card number, not PDF-checked) | C/U |
| Personas: ϕ differences all with CI including 0 | [50] | C |
| Kim's capability control removes the disagreement–gain effect on vision, 4 of 4 | [14] Kim 2026 (text only) | U/N |
| Debate, saturated pool: 10 of 11 fixes with a cap-truncated agent; gain depends on tie order; self-only keeps +1.7 of +3.5 on copies; CAPA rises only with peers | [16] Bertalanič and Fortuna (no card); [54]; [56] Goel | R + U |
| Debate, pool without frontier models (pre-registered): contested 0.801 → 0.870, ≈1.3 of 3.6 collected, coverage 1.000 → 0.925 | [54] Zhu | U |
| Single-model debate stays at the modal ceiling (+0.013); a mixed panel exceeds it (+0.061) | [26] (modal ceiling for sampling) | N/U |
| Judges: false-accept ϕ 0.38 on 54 wrong candidates; unanimity passes 1 of 54 at recall 0.85; panel never clearly above its best judge | [32] He (theorem); [57] Li | consistent with bound |
| Routing: no own run; per-set route collects 0.7 of ≤ 2.4 points (in sample) | [15] Chen (deployable routers ≈ 0.09 of the gap) | consistent |
| Decomposition: 0 of 7, Δ pass@4 0.00 [−0.29, +0.29]; +0.34 only where one pass is cut off | [60]; [61]; [62] | no effect detected (underpowered); C in tone |
| Equal tokens: best number of draws loses 0.3–31 points (68 at the worst); pays only on olympiad at B = 32,768 and GSM-Hard at B ≥ 8,192; nine copies add ≤ 2.7 | [63] Tran; against: [30], [65] | R + U; C |
| SWE-bench 175×500: majority 0.794 vs best 0.792, oracle +9.8; raw ρ above text, residual +0.05 after 2PL | [66] Ron; [47] Liu | R; N in scale |

| Our result | Closest prior work | Relation |
|---|---|---|
| Safety: misuse ρ 0.67 vs injection ρ 0.28, weakly shared (25/50 tasks) | [67] TAMAS; [68] | N (small n) |
| One-factor copula tight at k = 7 on SWE-bench and inside bands, breaks at 175; one eigenvalue above edge | [15] Chen; [53] (four eigenvalues) | U |
| Family residual after 2PL small but non-zero (+0.06… +0.11, ICC 8–14%) | [48] Kim 2025; [70] | U; C in size with [70] |

### 12.3 Cheap replications on the collected matrices (future work)

Not run for this paper; each needs only Python on matrices we hold.

- **[14] (Kim 2026):** share of k = 2–4 subsets where majority beats the best member (in sample and split-half) and collinearity of diversity with 1 − mean accuracy, on SWE-bench (top-30, k = 3) and the 733×7 corpus (35 triples). Done on public substitutes (pre-registered): a heterogeneous Condorcet model predicts held-out gain better than Kim's residualized double-fault (Spearman +0.81 vs +0.20 on MMLU-Pro), but its independent version is within +0.011 [0.005, 0.017], so dependence adds about 0.01.
- **[15] (Chen):** growth of the observed-to-predicted all-wrong ratio for k = 2…175 on SWE-bench; label audit of the 17 all-wrong MATH-500 problems.
- **[49], [57], [34]:** eigenvalue count vs Kish $n_{eff}$ on the judge matrix; dependence-aware k-of-n thresholds for OS-Harm breach screening; gain from a judge replaced only on one-vote queries.
- **[70]:** the copula's MAP floor $\Phi(\Phi^{-1}(1-\alpha) / (\sqrt{\rho}))$ against observed majority error as the number of members m grows, on SWE-bench.
- **[53], [26]:** within-model Marchenko–Pastur test and $\rho_b/\rho_w$ split on 16 DeepSeek samples.
- **[48] (Kim 2025):** agreement on identical wrong answers, regressed on company, architecture and accuracy.
- **[66] (Ron):** z-test of coincident failures against independence on SWE-bench.
- **[59], [50], [72], [56]:** majority vs number of models; 2 models vs 16 samples of one; lift law on 21 pairs; CAPA between models.

Not replicable on our data: CooperBench (no per-task logs), multi-agent verification (new verifier calls), ModelSwitch (needs samples at B/n).

### 12.4 Works to add to the reference list

**Verified** (card and corpus-manifest entry exist): [66], [63], [64], [61], [65], [58], [33], [34], [59], [54], [73], [72]. The disagreements with [33], [65], [30] and [50] need a reconciliation by operator, budget, verifier and member strength.

**To verify** (no card file and no manifest entry of their own; the id string appears only inside other papers' reference lists, or nowhere): [74] "More Agents Is All You Need", [75] "Debate or Vote", [76] Self-MoA, [77] Du et al. (debate), [78] mixture-of-agents, [16] Bertalanič and Fortuna, [79] "Two Calls, Two Moments" (no occurrence at all), and, cited from memory in the audit, [80] "Are More LLM Calls All You Need?", [81] "The Limits of Inference Scaling Through Resampling" and [82] (GCG jailbreak transfer).

## 13. What the data do not show

Each limitation below is one that a careful reader would raise; several of them change the strength of a headline statement, and we have tried to state the headline accordingly.

1. **The text corpus is saturated.** With the best member at 0.975 to 1.000 and the oracle within 2.3 points, a null on this corpus alone would be as much a property of the problems as of the operators. For voting the objection is answered: pools without the frontier models have 1 to 9 points of room (Section 5.4) and mid-ranked members of three public matrices have 14 to 32 (Section 5.5), and the vote collects none of it; both tests were decided after the fact. For debate and judges the non-frontier pool shows a small gain where the full pool showed none (Sections 6.4, 7.5), so the full-pool nulls for those two operators were partly a ceiling effect, and the ledger (Table 3.1) reports the non-frontier numbers as the operators' result. For the other operators the paper's claims about where ensembles do not pay rest on SWE-bench, τ-bench and the cost-matched sets, where there is room; the text corpus shows what happens when there is not.
2. **In-sample choices.** The best member, the tie order in debate and in the judge panel, the "strongest agent" who breaks ties, the unanimity stopping rule, the top-7 on SWE-bench (ranks 6 to 8 tie at 0.768 and the tie rule is not fixed), and the judge weights in the weighted vote were all chosen on the problems they are scored on. Where a held-out version was computed (best member, Section 5) it removed the effect; where it was not, the number is marked.
3. **The token cap.** Four of seven text models ran under 8,192 completion tokens. Every number that depends on truncated answers, which includes most of debate's gain and the olympiad self-consistency gain, would move under a larger cap; the direction is not known.
4. **One-sided label audit.** Only problems on which every model was wrong were re-read. Problems on which a model agreed with a wrong gold label remain, so 1 in 733 is a lower bound on joint failure, and accuracies are slightly over-stated for every model.
5. **Small denominators.** The olympiad set has 29 problems in the matrix and 49 to 60 in the samples; the judge experiments rest on 54 to 81 contested problems, half of them from one set; the decomposition null rests on 14 numeric problems of one model; OS-Harm has five agents, hence ten pairs, several with an empty cell of the 2×2 table. Intervals are given where they exist, and some headline gains (debate's, the self-only "half") have none.
6. **Leaderboard submissions are not independent runs.** The 175 SWE-bench entries contain 74 distinct scaffolds and 146 distinct scaffold–model pairs; the same model appears three times in the top 7. Removing duplicates leaves the result in place ($n_{\text{eff}}$ 1.43, majority 0.784 below best 0.792) but the panel is still whatever the leaderboard happened to hold.
7. **Partial verifier: two data sets, both released by others.** Section 7.7 measures one model verifier (R2E-Gym) and one execution-based rule (an agent's own tests, SWE-rebench). The R2E analysis was a first look before its pre-registered part; one agent and one verifier per set; the test verifier available is the solver's own, which is lenient ($\alpha$ 0.69); the strict test verifier that the formula favours exists in the data only with the reference fix. Our own runs (Table 7.6) add a judge and two test writers of one model on 200 contested issues, chosen on the hidden labels, with at most five tests per issue; the only strict execution-based verifier in any data set uses the reference fix. Unmeasured: a verifier with a small false-accept rate that needs no hidden information, and the verifier's independence from the solver (the residual error correlation between model families on SWE-bench after the difficulty factor is at most 0.09, so that lever is small).
8. **One model in the cost-matched study** and roughly sixty problems per set; nominal prices; no run of a single model given more than its natural budget; the verifier's own cost unpriced.
9. **No open-ended tasks.** Every problem has a checkable answer. On writing, design or planning the oracle is undefined and the whole accounting changes; the paper says nothing about that regime.
10. **Missed pre-registered predictions.** The four predictions written for the non-frontier debate and judge run (Section 6.4) all failed, in the direction of a larger effect than the full pool had suggested; they are reported as misses, and the resulting numbers are exploratory. Two SWE-bench predictions also missed, toward more correlation (Section 11.2).
11. **Multiple comparisons.** Across Sections 5 to 9 several dozen operator-versus-member comparisons are reported without correction; the one that reaches $p < 0.05$ is debate under a fixed tie order, and it does

not survive the strongest-first order ($p = 0.06$).

12. **External claims are read, not re-run.** The orchestrator results of Section 8 come from the systems' own reports and are compared on the gap "oracle minus best" where the reports allow it; parity claims against provider-reported baselines cannot be checked by us.
13. **Provenance of the analysis.** Several results were obtained in exploratory passes and revised after audits; the published notes this paper draws on [83, 84, 85, 17, 86, 87, 88, 89, 90] carry those revisions as dated addenda, and the numbers here follow the audited versions.

## 14. Conclusion

An ensemble of language models can only redistribute what its members already know how to do; the oracle is its ceiling, and the distance from the best member to that ceiling is the whole prize. Operators that only rearrange the members' answers collect none of it (voting) or a small minority of it (routing, by published tables) on any data set we measured, whether the prize is two points on a saturated corpus or thirty on a mid-ranked band of agent submissions, because the members fail on the same hard problems: three independent methods say that their shared error is the difficulty of the item and almost nothing else. Operators that add computation collect a few points, and only under conditions we can name: a single draw well below 0.9 and a budget around ten times one answer for repeated sampling; members whose errors are not already shared for debate, which then pays in coverage; a judge that solves better than the solvers it reads for selection. Decomposition lifts a model that runs out of tokens and nothing else.

The practical reading is a rule rather than a verdict. Before building an ensemble, measure the oracle of the pool and the effective number of votes; if the first is close to the best member or the second is close to one, no rearranging operator will pay, and the money is better spent on a larger budget for the best member. If the oracle is far above the best member and the members disagree, the operator to build is a selector with a verifier that is independent of the solvers, and the two numbers to measure are that verifier's false-accept and false-reject rates, per task. On released agent candidates (Section 7.7) a model verifier with no small-false-accept operating point collects half of a 32-point prize at two candidates and a third at sixteen; tests written by a model, whether the solver's own, from the issue text or after reading the code, accept 39 to 69% of wrong patches and so supply recall rather than a small false-accept rate; a per-task two-rate formula reproduces all of it; and the best deployable rule, such tests as a gate before a ranking judge, collects 0.28 of the prize at eight candidates. In one sentence: a verifier's two error rates, measured per task, predict what a selector collects; on agent tasks the prize goes to a verifier with a small false-accept rate, and neither model judges nor model-written tests, blind or reading the code, supply one, so the best deployable rule found collects 0.28 of the prize at eight candidates by gating on tests and ranking by a judge. The verifier the formula asks for, one with a small false-accept rate that does not use the hidden tests, is the one still to be found.

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